MES4-Carrier-EC · design review package

Input for review R3 · schematic Rev A, 28 Oct 2026 (branch `mes4-gpio-standard`) · Monkey Robotics · Aaron Raj Bhatti

What's in this package: (1) what changed since review R1 and the status of every R1 finding, (2) whole-board LTspice simulations (review item M4), grouped by schematic sheet, (3) the updated 11-page schematic, appended after this report.

Since the R2 package (27 Oct): U10 is now the 1″ × 1″ YLPTEC URB2405YMD-20WR3 (C5369777) instead of the 2″ × 1″ URB2405LD-20WR3, same 5 V 4 A / 20 W isolated rating and 50 V 1 s surge, with a 100 µF / 16 V output bulk C38 (CX RVT1C101M0605, C81993) per its datasheet; AUX1 re-run with it (pass). The FE-URT-2 servo driver now mounts on four YIYUAN SMTSOM320BTR M3 × 2 mm SMD standoffs (H6–H9, C5301780) as on the MES3 carriers, and a simpleRTK4 Dual GPS mount has three SMTSOM365BTR M3 × 6.5 mm SMD standoffs (H10–H12, C5301796). No other circuit changes.

1. Changes since review R1
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1 · Changes since review R1

Sheet (page)	Change	Why
Relays 1-2 / 3-4 (5, 6)	U2–U5 **INA226 → INA236AIDDFR** (LCSC C5443912, TSOT-23-8). Addresses now 0x40 / 0x41 / 0x42 / 0x43 via A0 = GND / VS / SDA / SCL. A1 and VBUS are gone; the bus voltage is read on IN–. Same CONFIG 0x4327, CAL 1707, MASK/ENABLE 0x8001 and ALERT limit; bus LSB 1.6 mV, Device ID 0xA080	M1: 50 V abs max vs the INA226's 40 V (SMBJ30A clamps up to 48.4 V). The 85 V INA238 / INA228 had 0 LCSC stock
AuxPower (9)	**J15 servo 24 V: pin 1 = +24 V, pin 2 = GND_24V**, like every other power terminal	M3. The Harvester CN5 cable gets re-pinned
AuxPower (9)	**R38** (servo RX series) **100 Ω → 1 kΩ** (LCSC C11702)	L4: limits FE-URT-2 back-feed into PA3 with the module off
AuxPower (9), Main (2), Power (4)	IMU note: **3DM-CV7-AHRS on the C-Series Connectivity Board** (6212-3009). J14 9 V → board Vaux only, never Vpri; not with the USB cable at the same time. IMU load in the power budget ≈ 0.02 A at 24 V	The IMU is now known; Vpri (3.2–5.2 V) would be destroyed by 9 V
Relays 1-2 (5)	Diagnosis note: "current present" threshold **5 mA → 3 mA**; INA236 differences listed	L2: the LED alone draws 5.1–6.7 mA over 21–29 V

Power (4)	F1 / Q1 notes use the real worst case: 13 A, Q1 ≈ 2.2 W, size the drain copper for 2.2 W	L6
LoadCell (8), Main (2)	Silkscreen notes: filled box **"5V LOAD CELL – NEVER 24V"** at J12 / J13, placed away from the 24 V terminals; **"CAN – NOT ETHERNET"** at J4	M2 (silkscreen instead of a keyed connector, by choice: JLC doesn't stock the mating plugs) and L5
Footprints	**J4 RJ45 1×4 rebuilt from the HCTL drawing**: ports 14.0 mm apart (was 13.97), Ø1.5 shield holes (was Ø1.7), shield and peg positions per the drawing. U1 / U9 converter footprints checked against the JLC library and the YLPTEC drawing; U10 (URB2405YMD-20WR3) against the JLC library and the Mornsun drawing	L9, and the custom-footprint check before routing
sim/ (new)	16 LTspice benches covering every sheet with an analog or power circuit, on the real supply chain (sections 4–12)	M4

> Unchanged since R1: the GPIO map (no coil feedback; PA0/PA1 winch, PA2/PA3 servo USART2), the servo TX pull-up R37, the Winch_48V net class and isolation notes, the block diagram and the page 11 finger map. All of these were already in the R1 schematic.

2 · Status of the R1 findings

ID	Finding	Status	How
M1	INA226 40 V vs the SMBJ30A clamp	Fixed	INA236 (50 V). PWR2: the bus peaks at 40 V at D9's rated pulse
M2	24 V miswire on the load-cell terminals	Mitigated	Silkscreen box and placement away from the 24 V terminals (no keyed connector, by choice)
M3	J15 polarity	Fixed	Pin 1 = +24 V
M4	No simulation evidence	Done	16 benches, sections 4–12
L1	Relay LED resistor power	No change needed	R14 / R15 / R22 / R23 are already 1206 (0.25 W)
L2	5 mA threshold	Fixed	3 mA in the diagnosis note (firmware)
L3	Servo return through logic GND	Layout	Layout checklist: J15, J17 and NT1 together, away from the load cell
L4	PA3 back-power	Fixed	R38 = 1 kΩ; AUX3 shows 8.8 ns edges at 1 Mbit/s
L5	RJ45 looks like Ethernet	Silkscreen	"CAN – NOT ETHERNET" on the board and the panel
L6	Stale Power notes	Fixed	13 A / 2.2 W

L7	3 A PTC derating	Open	AUX1: F11 sits at its 3 A hold if the screen draws 3 A. Measure the screen; 4 A-hold PTC if needed
L8	Screen on 5 V only	Bench	Bench test at full brightness
L9	DC/DC pin numbering	Closed	U1 / U9 / U10 match the JLC library; U10 (URB2405YMD-20WR3) also the Mornsun drawing

3 · Open items for the reviewer

- **F11 (screen PTC):** keep 3 A hold, or fit 4 A now? It depends on the ARZOPA Z1C's real current (AUX1, L7).
- **Supply limits:** the pack's full-charge voltage (7S Li-ion 29.4 V or 8S LiFePO4 29.2 V) against the AB20 variant (A/B/C rated 30 V, D rated 60 V).
- **Datasheet checks: done, all pass.** URB2405S-6WR3, URB2409S-10WR3 and URB2405YMD-20WR3 all allow a 50 V input surge for 1 s (PWR2 peaks at 40 V). Q1 AGM60P90D I_{DM} −360 A (inrush 100 A). F1 0451015 melting I^2t 97.8 A^2s (inrush 0.95 A^2s).
- **Hold-up:** 4.0 ms on total power loss (PWR6), to a 9 V cut-off. The YLPTEC datasheets give an undervoltage shut-off of 5.5–6.5 V, so the real figure is a little longer. Enough for glitches; add bulk on +24V only if a clean shutdown is wanted.

4 · Simulation summary

> LTspice 17.2.4. Benches are in `MES4_Carrier_EC/sim/` and named by sheet: PWR = Power, RLY = Relays, SNS = Sensors, LC = LoadCell, AUX = AuxPower, IMU = AuxPower → dev kit, WCH = Winch, MAIN = Main. Each section below lists the sheet, the parts simulated and the bench file.

Group	Sheet (page)	Bench	Test	Result
Power	Power (4)	PWR1	XT30 hot-plug, full pack (29.4 V) via the AB20, 2 m cable	Pass
Power	Power (4)	PWR2	12.4 A 10/1000 µs pulse (D9 rating)	Pass
Power	Power (4)	PWR3	IEC 61000-4-5 0.5 kV surge	Not applicable
Power	Power (4)	PWR4	Battery connected backwards	Pass
Power	Power (4), fed by the AB20	PWR6	Line changeover and hold-up	Pass
Power	Power (4) + Main (2)	PWR5	+5V_24 / module +5V under load steps	Pass
Relays	Relays 1-2 / 3-4 (5, 6)	RLY1	SRD coil driver on / off	Pass
Relays	Relays 3-4 (6)	RLY2	Solenoid output switched off	Pass
Sensors	Sensors (7)	SNS1	PC817 input, 20 V sensor, CTR 50 %	Pass
Load cell	LoadCell (8)	LC1	ADC input filter, AC response	Pass
Aux power	AuxPower (9)	AUX1	U10 start-up into the screen	Pass, check F11
Aux power	AuxPower (9)	AUX2	3 A servo stall step	Pass
Aux power	AuxPower (9)	AUX3	Servo UART, 1 Mbit/s	Pass
Winch	Winch (10)	WCH1	IM03GR coil driver on / off	Pass
IMU	AuxPower (9) → J14 → dev kit	IMU1	SW2 switches 9 V into the CV7 dev board	Pass
Main	Main (2) + Relays (5, 6)	MAIN1	I²C1 at 400 kHz, ALERT	Pass

> Everything still to decide or check is collected in section 3.

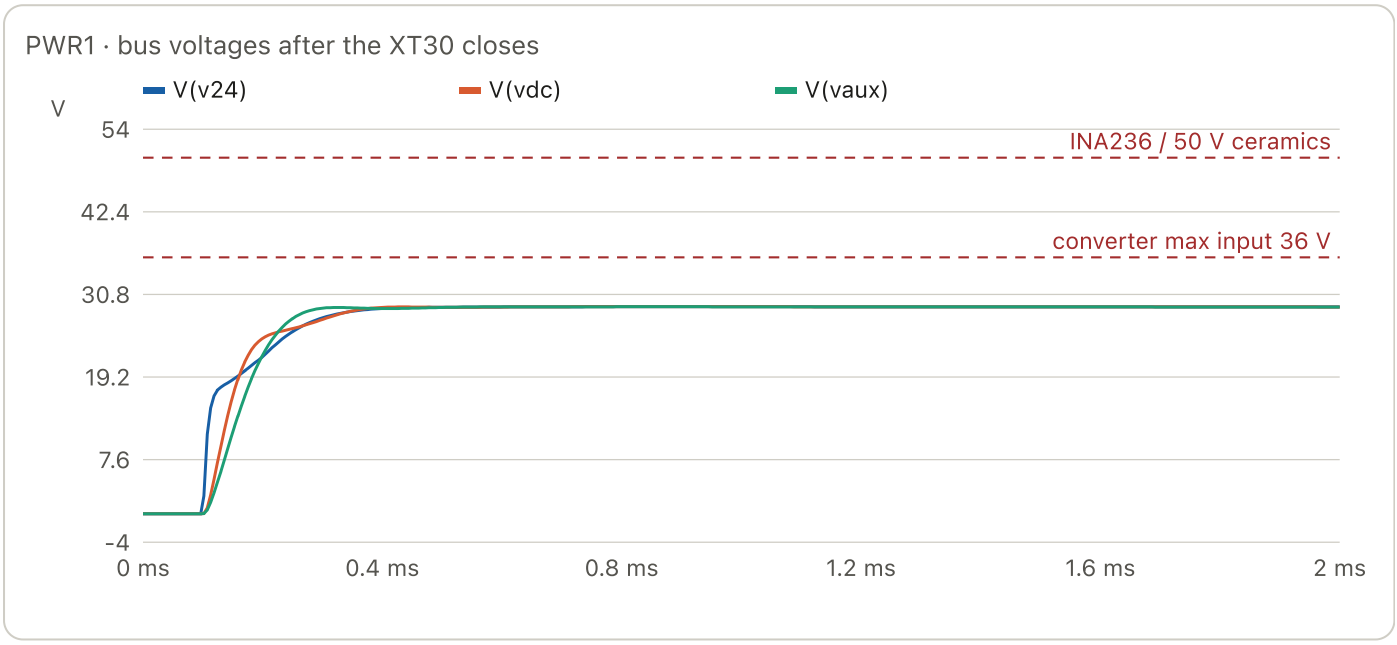
5 · Power · sheet Power (page 4)

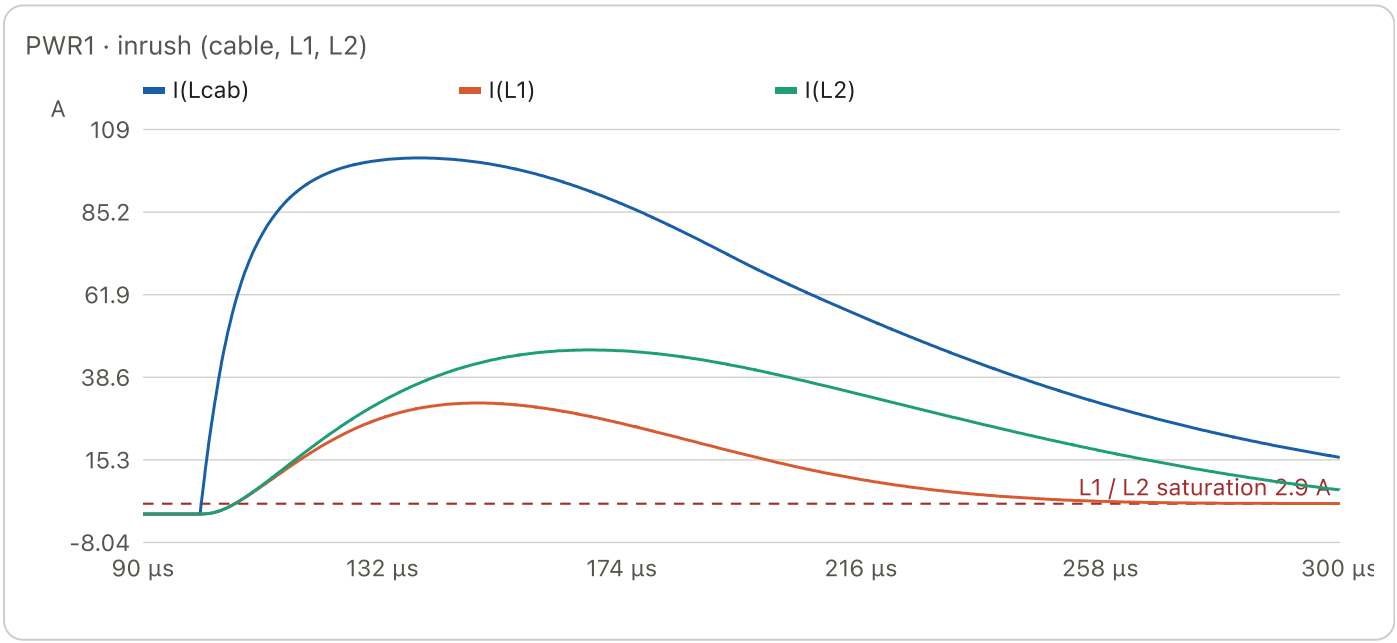
Supply into J5 (your system): 48 V main battery → Mean Well **DDR-480C-24** (set to 24 V; adjustable 24–28 V; OVP 28.8–35 V; 4 kV isolated; 20 A) → **AB20** line 1. 24 V 12 Ah Li-ion pack → **XY-CD63L** discharge monitor → AB20 line 2. The AB20 is a diode-OR ("zero-delay", anti-backflow): the higher input feeds the carrier. Model: `source_ab20.inc`.

Parts: J5 XT30, F1 15 A, Q1 AGM60P90D + D8/R6, D9 SMBJ30A, C1–C3 · L1 + C4/C5 → U1 URB2405S-6WR3 (+24V_DC) · L2 + C26–C30 → U9 / U10 (AuxPower) · F9 + C31/C32 (servo). Shared model: `common.inc`.

PWR1 · hot-plug (`pwr1_hotplug.net`)

The XT30 closes on the AB20 output with the pack fully charged (29.4 V, above the DDR's 24 V, so the pack feeds the carrier), through a 2 m cable (2 µH, 33 mΩ). +24V rises cleanly to 29.1 V, with no overshoot on any branch, so D9 never conducts. Charging about 450 µF pulls a current pulse peaking at 100 A and lasting about 0.1 ms (F1 I^2t 0.95 A^2s), so the XT30 will spark. L1 and L2 saturate briefly (31 / 46 A in this linear model). That's harmless: Q1 AGM60P90D is rated −360 A pulsed (AGMSEMI datasheet), and F1 0451015's nominal melting I^2t is 97.8 A^2s (Littelfuse), about 100× the simulated 0.95 A^2s.

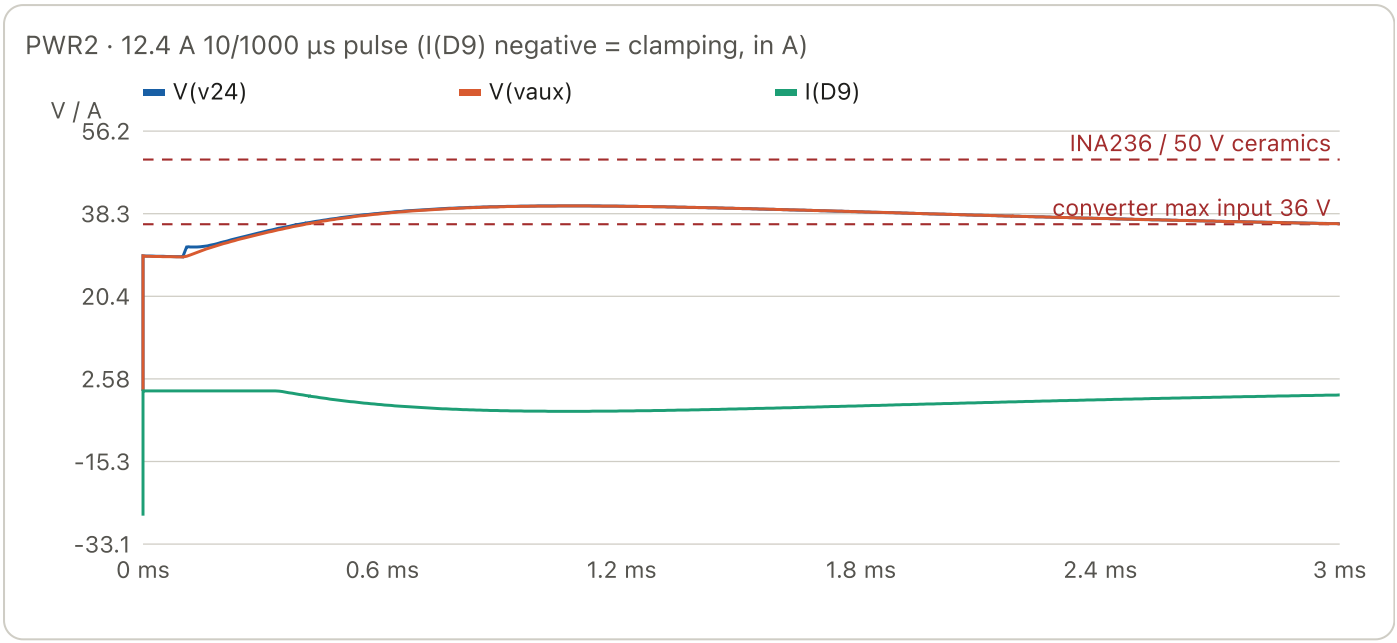




PWR2 · surge at D9's rating (`pwr2_tvs_pulse.net`)

This is the design-limit case: both AB20 inputs are gone (bus held at 29.2 V by a weak source) and a 12.4 A, 10/1000 µs pulse arrives on the cable. With the AB20's diodes blocking any return path, this is also what servo back-EMF would face. The bulk capacitors take most of it: the bus peaks at 40.0 V and D9 takes 4.4 A and 0.31

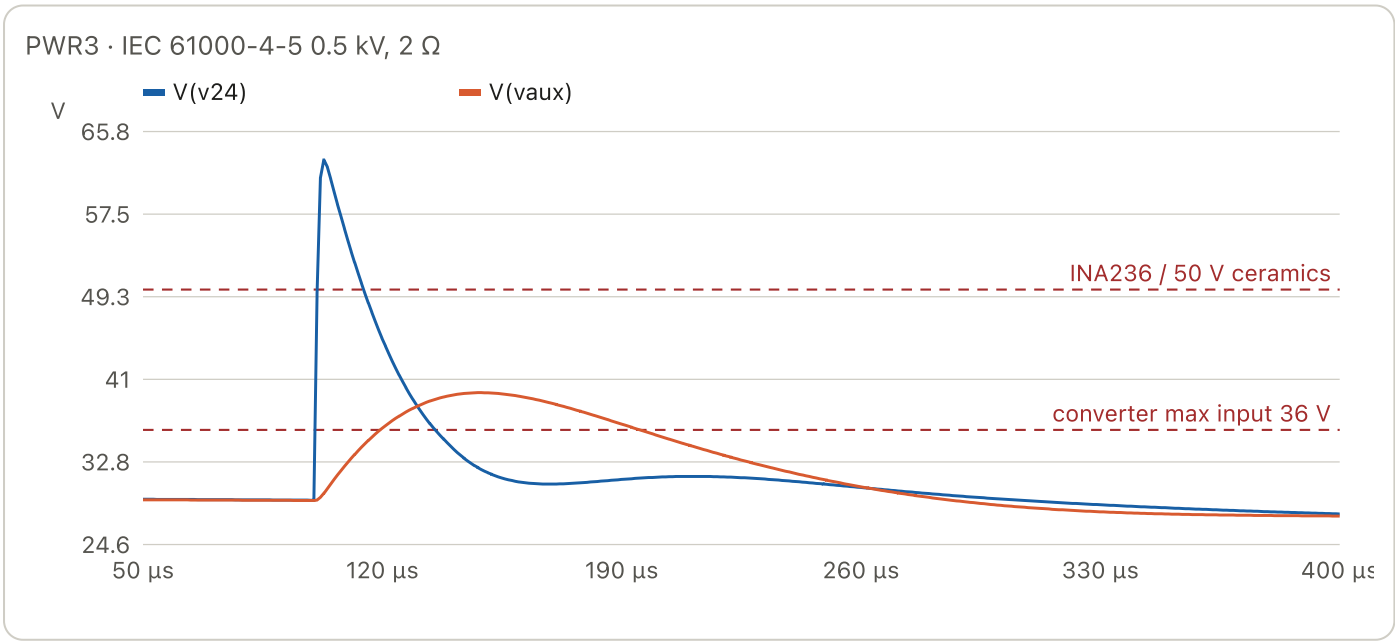
J of its ≈ 0.87 J rating. That leaves 10 V of margin to the INA236. U1, U9 and U10 see 40 V for about 1 ms: above their 36 V operating range, but inside the 50 V for 1 s input surge rating on all three YLPTEC datasheets.



PWR3 · IEC 61000-4-5 surge (`pwr3_iec_surge.net`) · not applicable

Not applicable to this system: the carrier's 24 V comes from the isolated DDR-480C-24 and an internal pack through the AB20, on short wiring inside the robot. The run is kept for reference, in case the bus ever gets an external charger port or a long harness.

0.5 kV line-to-line, 1.2/50 µs, 2 Ω, through an 18 µF coupling capacitor, with the battery behind a 1.5 mH decoupling inductor. D9 conducts 27 A and the bus reaches 62.9 V, above the INA236's 50 V and the 50 V ceramics. +24V_AUX stays at 39.7 V behind L2. This surge is normally required only for long or external DC power cables. If the bus ever goes off the robot (a charger port, a long harness), fit an **SMCJ30A** (1500 W, same standoff, Vc 48.4 V at 31 A, about 47 V here).

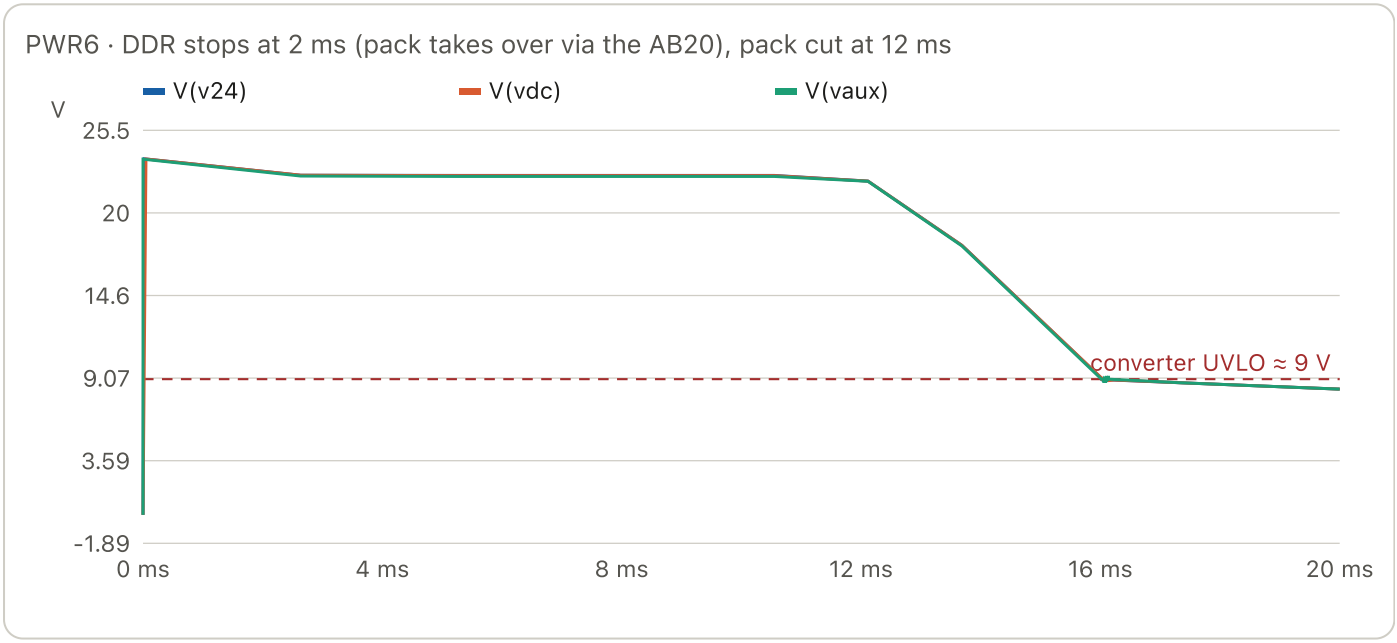


PWR4 · reverse battery (`pwr4_reverse.net`)

The battery is connected as −29.2 V. Q1's body diode is reverse-biased and its gate stays off, so +24V stays between −4 mV and +2 mV and only 0.12 mA of leakage flows. D9 sits after Q1, so it never sees the reverse voltage. Pass

PWR6 · line changeover and hold-up (`pwr6_changeover.net`)

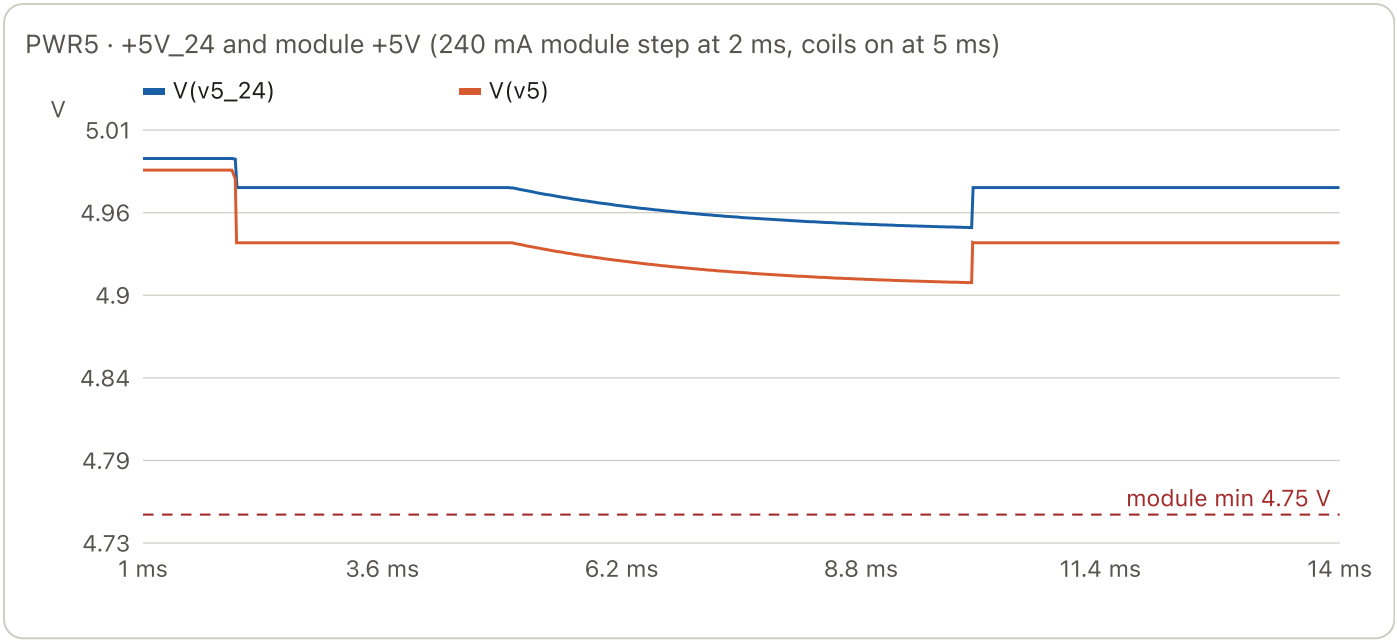
The DDR on line 1 carries the load at 24 V (the carrier sees 23.6 V after the AB20 diode), with the pack part-discharged at 23.0 V. At 2 ms the DDR stops (48 V battery below its 33 V UVLO, or an OVP trip). The pack takes over through the AB20 with no gap: +24V simply steps to 22.5 V. At 12 ms the XY-CD63L cuts the pack as well. With both sources gone, the carrier keeps running on its own bulk capacitance for **4.0 ms** (screen at full load) before U1's input falls to 9 V and the module loses 5 V. That's enough to ride through glitches, not enough to shut down cleanly. If you ever want a clean shutdown on total power loss, add bulk capacitance on +24V (2,200 µF would give about 20 ms). Also, the DDR alone can never put more than 35 V on the bus (its OVP), which is inside the converters' 36 V and the INA236's 50 V.



PWR5 · 5 V rails (`pwr5_5v_rail.net`, sheets Power + Main)

Parts: U1 output → C6/C7 (+5V_24) → F2 1.1 A PTC → FB1 → C8/C9 → module fingers 20/22 (module 10 µF + SMF6.0A); relay K1–K4 and winch K5 coils on +5V_24; load cell excitation via FB2.

The module steps from 40 mA to 240 mA at 2 ms, then four relay coils and one winch coil switch on together at 5 ms (0.27 A after 5 ms, rising to 0.31 A). The module's +5V stays between 4.91 V and 4.99 V, inside its 4.75–5.5 V window, and +5V_24 never drops below 4.95 V.



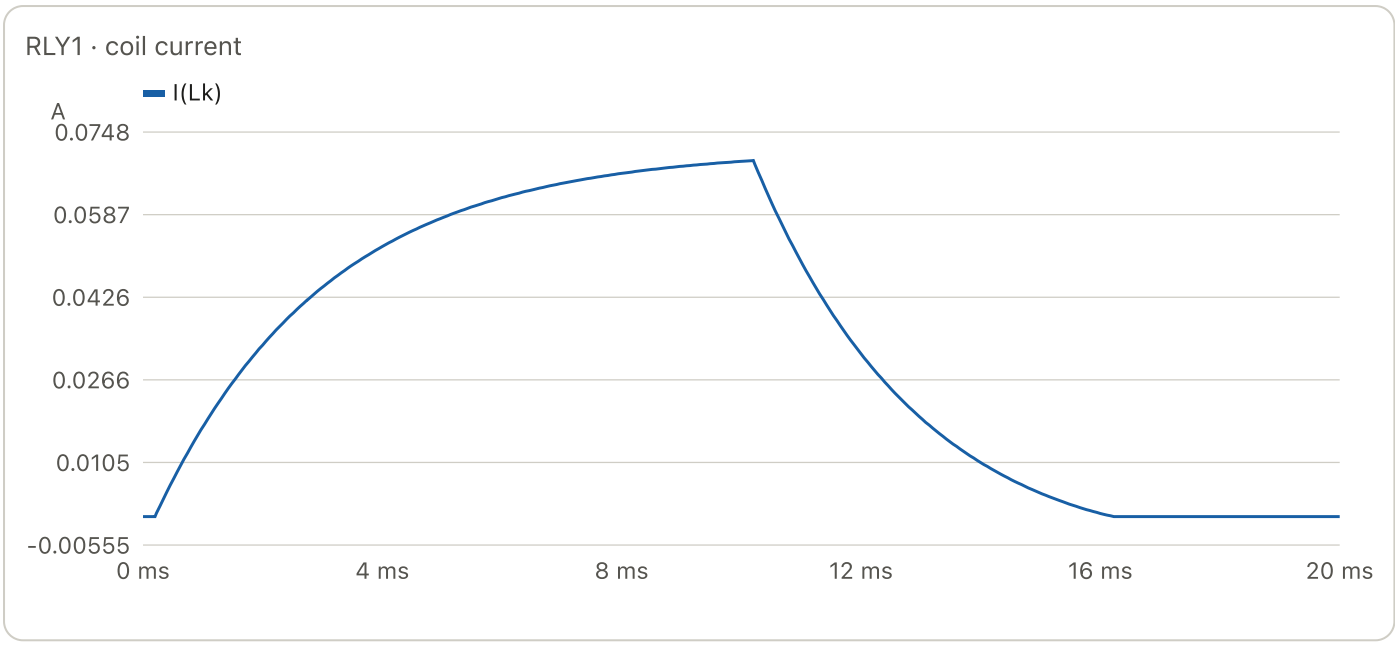
6 · Relays · sheets Relays 1-2 / Relays 3-4 (pages 5–6)

Parts per channel: GPIO → R9 100 Ω (R10 100k pull-down) → Q2 AO3400A → K1 SRD-5VDC-SL-A coil, D10 1N4148W flyback; contact → F3 3 A PTC → J6, D12 SS34 load flyback, LED4 + R14; R8 30 mΩ shunt → U2 INA236. Same for channels 2–4.

RLY1 · coil driver (`rly1_coil_driver.net`)

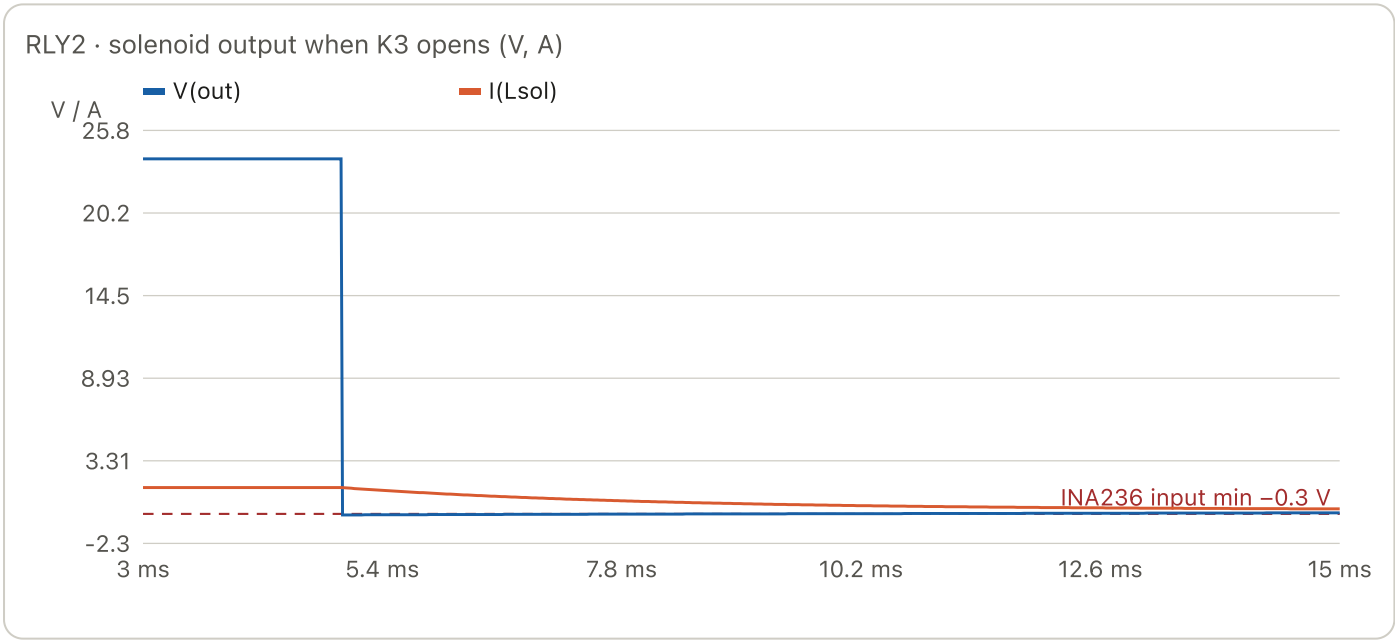
3.3 V GPIO on at 0.2 ms, off at 10.2 ms. The gate rises in 0.28 µs, and the coil draws 69 mA. At turn-off, D10 clamps the drain at 5.8 V (the AO3400A is rated 30 V). The coil current takes 3.9 ms to fall to 10 mA, which delays the relay's release a little. That's normal with a plain diode flyback and fine for lamps and the solenoid.





RLY2 · solenoid switched off (`rly2_solenoid_off.net`, channel 3 battery stopper)

A 24 V, 1.5 A solenoid (16 Ω, 50 mH assumed) is on when K3 opens at 5 ms. D16 SS34 clamps the output at −0.36 V and carries 1.49 A (rated 3 A). The current decays to 10 % in 6.9 ms. The INA236's IN− sits on the supply side of the contact and stays at 23.96 V, so the flyback never reaches its −0.3 V input limit.

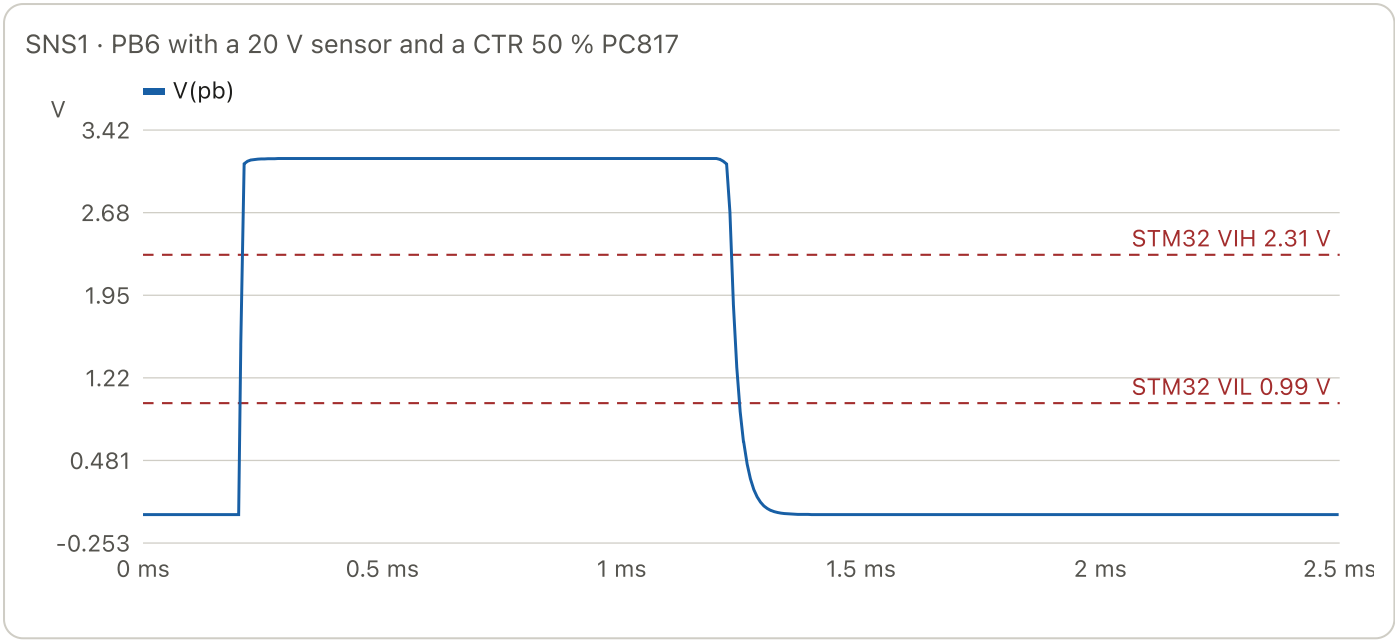


7 · Sensors · sheet Sensors (page 7)

Parts: J10 → F7 100 mA PTC → sensor; sensor output → R24 10k → U6 PC817 (D18 reverse clamp) → PB6 with R25 10k pull-down, collector on +3V3_MOD. Sensor 2 (PC15) is identical.

SNS1 · PC817 input (`sns1_pc817.net`)

This is the weak case: the sensor output is only 20 V (low battery) and the PC817 has its minimum CTR of 50 %. The LED draws 1.84 mA. PB6 goes HIGH to 3.17 V, well above the STM32's 2.31 V threshold, with about 3× margin on CTR. Rise is 4 µs and fall is 34 µs, much faster than any proximity sensor switches.

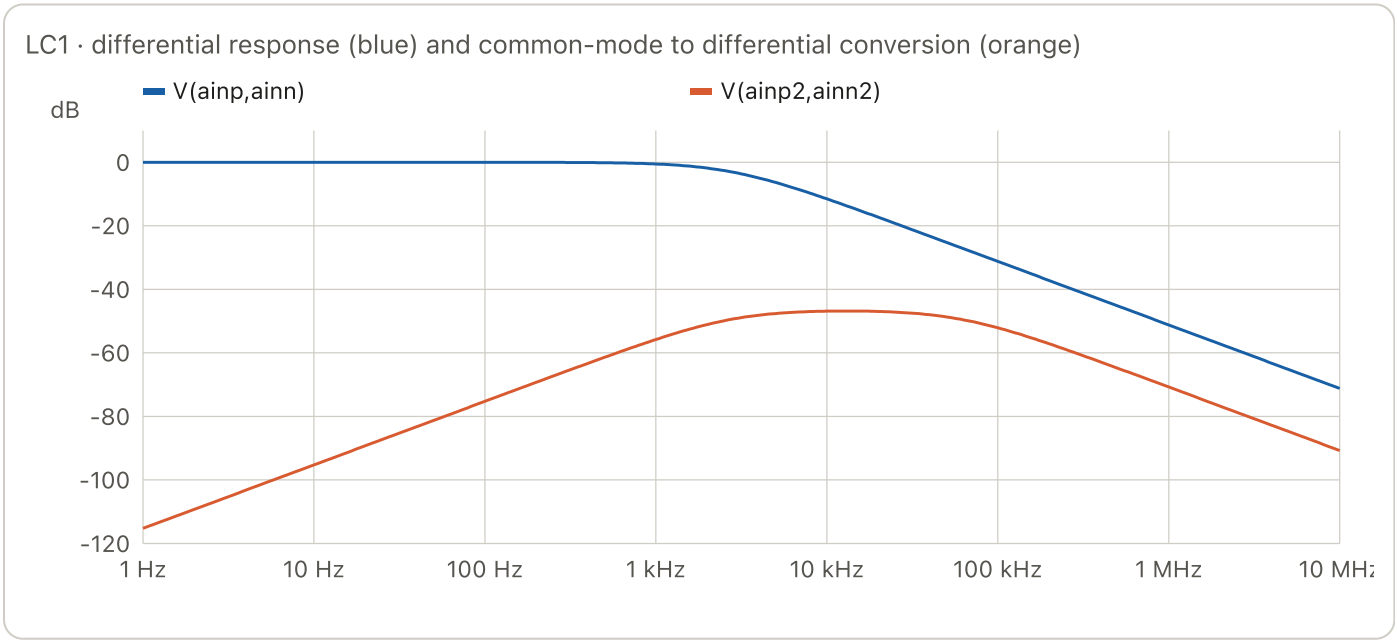


8 · Load cell · sheet LoadCell (page 8)

Parts: J12 SIG → R28/R29 100 Ω → C17 100 nF differential + C16/C18 10 nF to GND → D20/D21 BAV199 clamps → U8 ADS1232 (gain 128, 10 SPS). Source: two SOPLF-102 cells in parallel (700 Ω each → 350 Ω).

LC1 · input filter (`lc1_input_filter.net`, AC)

The differential signal is flat in the passband (−0.001 dB at 50 Hz) and rolls off at −3 dB at 2.76 kHz, including the cells' 350 Ω source. Common-mode noise converts to a differential error only through component mismatch. With C18 10 % low, that's −56 dB at 1 kHz, worst ≈ −47 dB around 20 kHz and −71 dB at 1 MHz. The ADS1232's 10 SPS digital filter removes what's left at those frequencies. The C0G capacitors keep the mismatch small and stable.

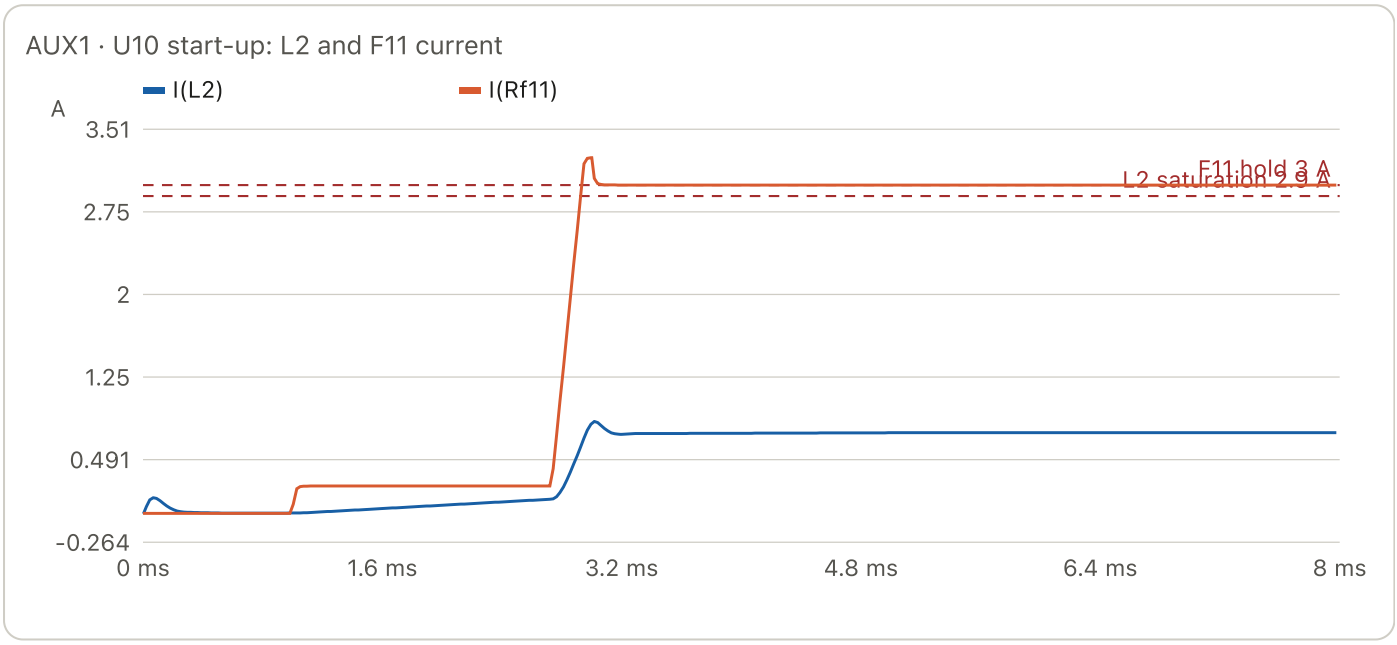


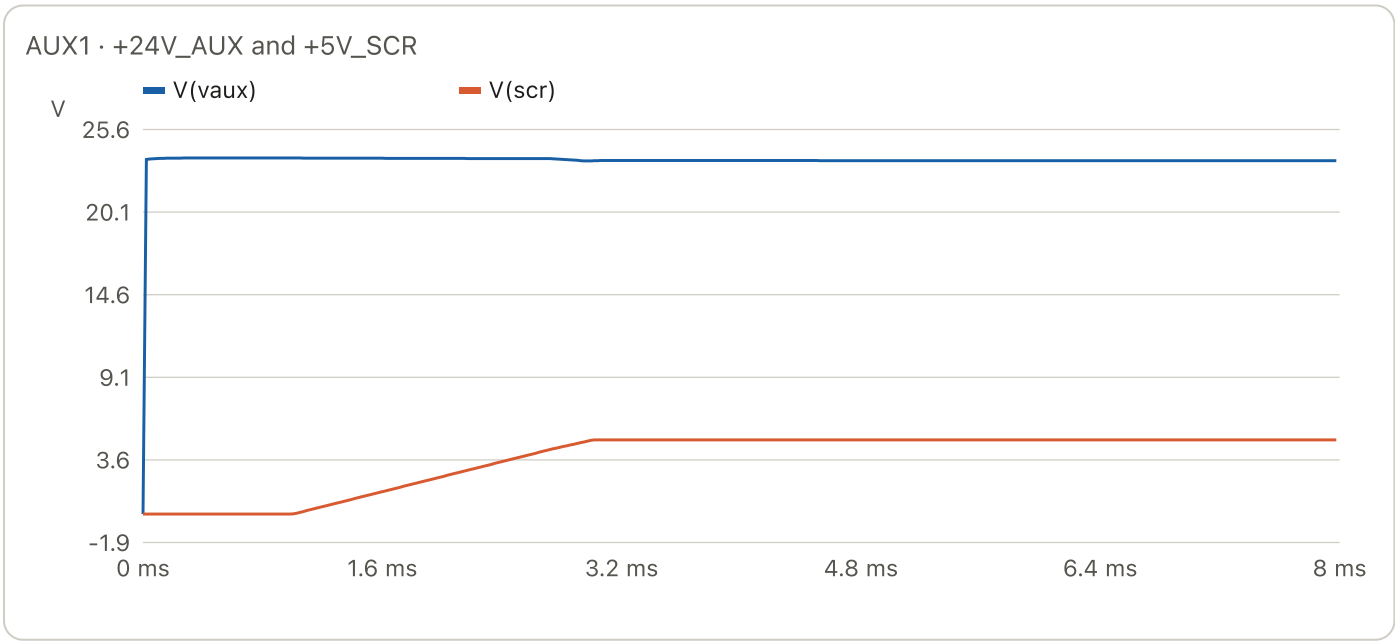
9 · Aux power · sheet AuxPower (page 9)

Parts: L2 + C26/C27/C28/C30 → U9 URB2409S-10WR3 (IMU 9 V) and U10 URB2405YMD-20WR3 (screen 5 V) → C33/C34/C38 → F11 3 A PTC → J16 USB-C · F9 3 A PTC + C31/C32 → J15 servo 24 V · servo UART J17: R39 100 Ω / R38 1k series, D23/D24 ESD, R36/R37 10k pull-ups.

AUX1 · U10 start-up (`aux1_u10_startup.net`)

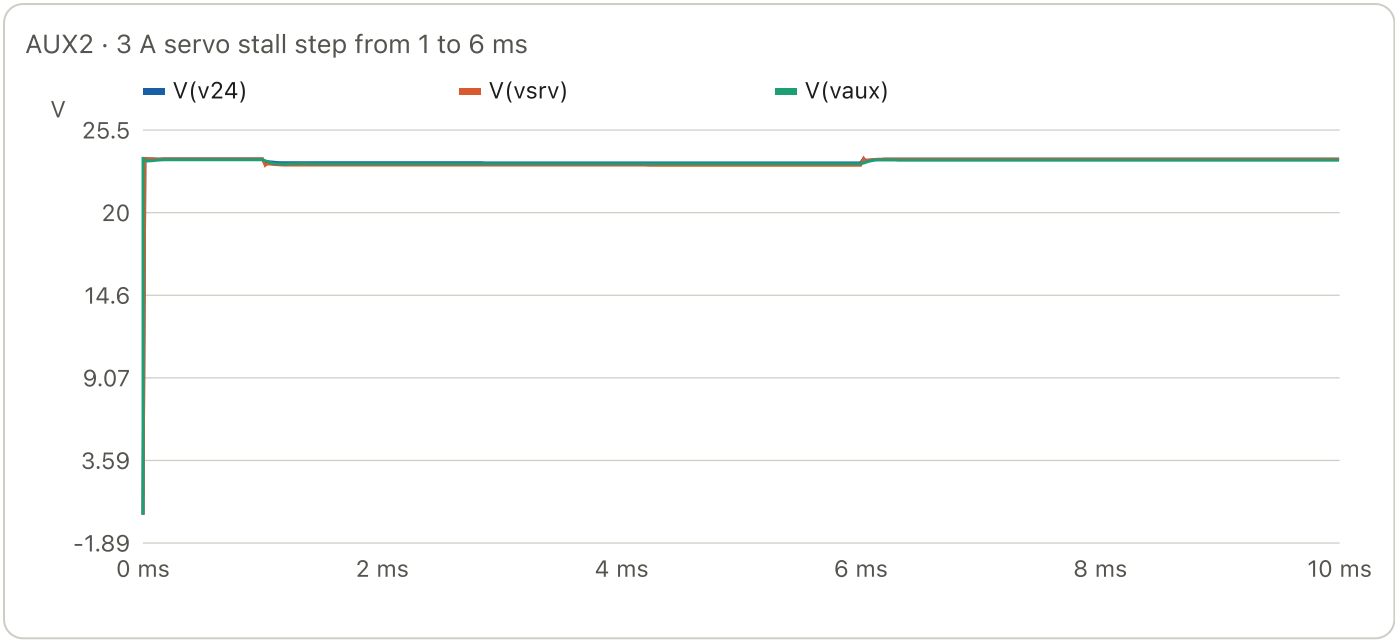
U10 is now the 1″ × 1″ URB2405YMD-20WR3 (9–36 V in, 40 V max, 50 V for 1 s; 5 V 4 A; 88 % minimum efficiency; over-current trip 110–190 % of 4 A, hiccup). With the bus up, it soft-starts its 5 V output in 2 ms (fast-case assumption; the datasheet's typical start-up is 10 ms, which is gentler) into C33/C34, the new 100 µF C38 (the datasheet's Cout), F11 and the screen, which pulls the full 3 A. U10 peaks at 3.58 A while charging the output capacitors: 0.8 A under the lowest possible over-current trip (4.4 A), so it can't go into hiccup at start-up. It then runs at 3.0 A (15 W, 75 % of its 20 W rating) with +5V_SCR at 4.94 V, inside its ±3 % tolerance. The total output capacitance (≈ 240 µF with the screen) is far below its 10,000 µF limit, and at 75 % load it needs no heatsink up to about 70 °C. L2 carries 0.84 A in total (U9 with the IMU + U10), far below its saturation current, and +24V_AUX dips only to 23.5 V. F11 peaks at 3.25 A and sits at 3.0 A, exactly its 23 °C hold current. In a warm enclosure (≈ 2.2 A hold at 60 °C) it would trip if the screen really draws 3 A.





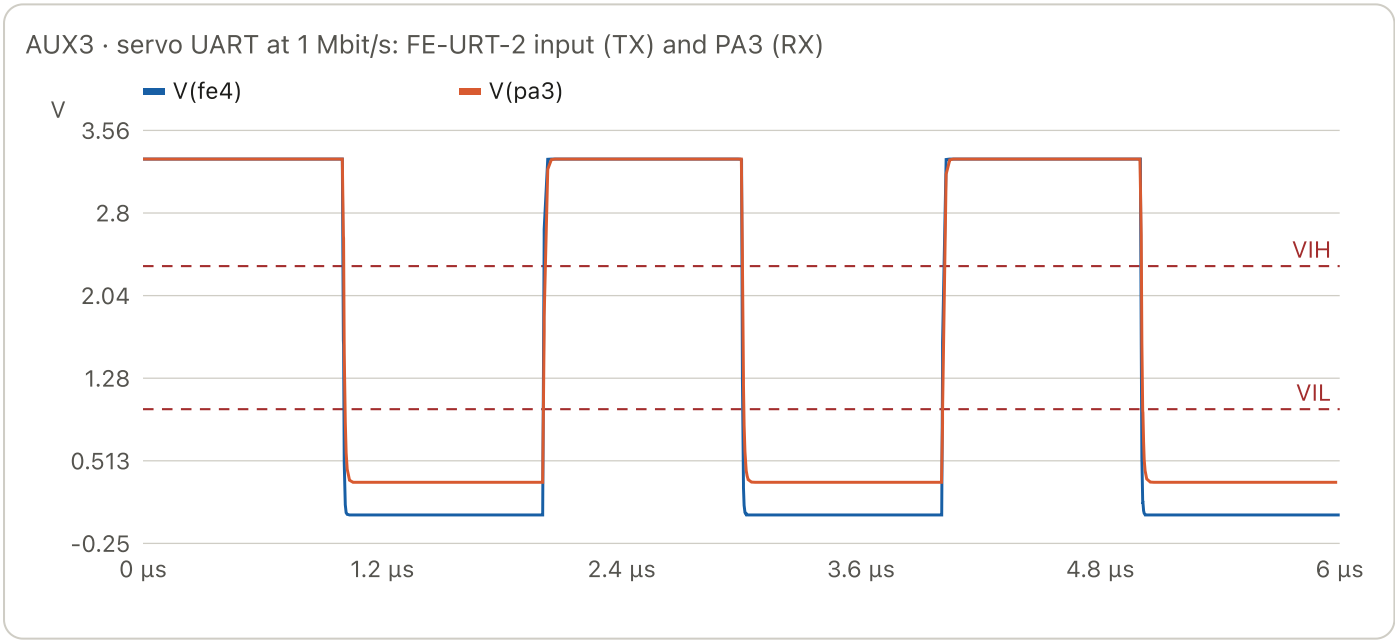
AUX2 · servo stall step (`aux2_servo_step.net`)

The servo draws 3 A for 5 ms (stall), supplied through the AB20 from the DDR at 24 V over a 2 m cable. +24V_SRV dips to 23.19 V and the main +24V to 23.32 V. The relays, INA236s and +24V_AUX (23.27 V) barely notice.



AUX3 · servo UART (`aux3_servo_uart.net`)

1 Mbit/s (Feetech default) over 0.3 m, with the FE-URT-2 at 3.3 V levels. TX edges at the FE-URT-2 rise in 4 ns. RX edges at PA3 rise in 8.8 ns through the 1 kΩ R38: both under 1 % of the 1 µs bit. PA3's LOW level is 0.31 V (1k against the 10k pull-up), under the STM32's 0.99 V threshold.



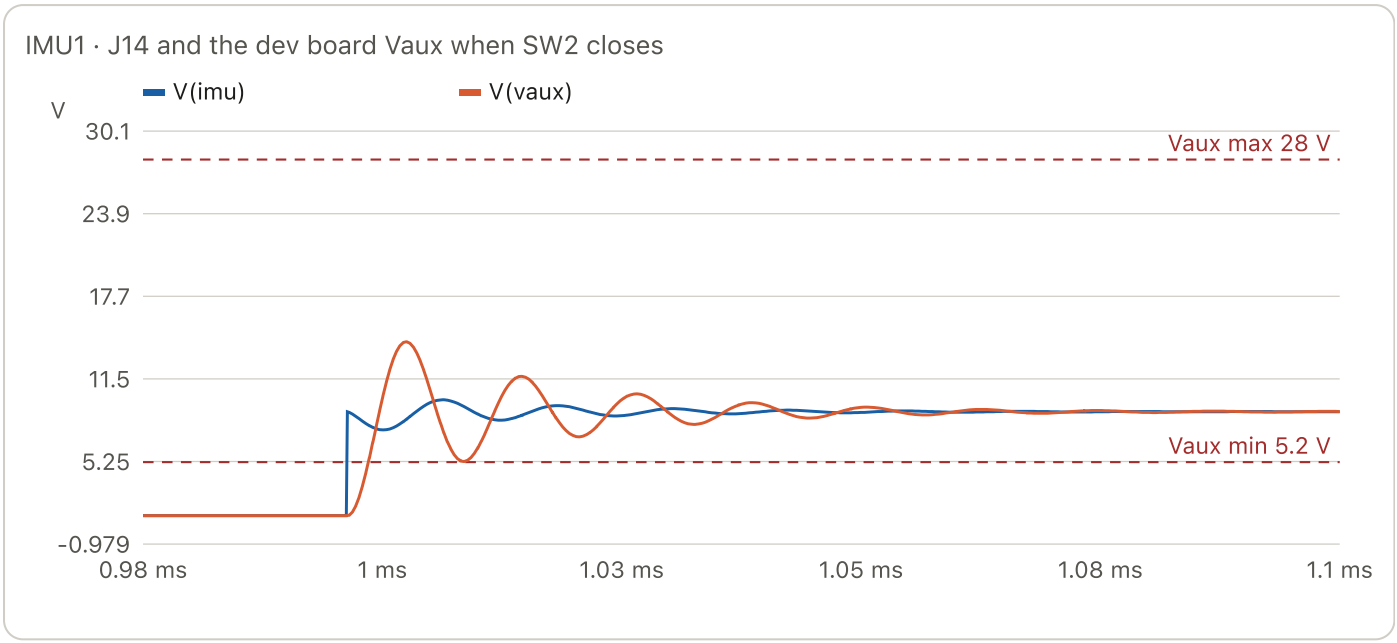
10 · IMU · sheet AuxPower (page 9) → J14 → MicroStrain dev kit

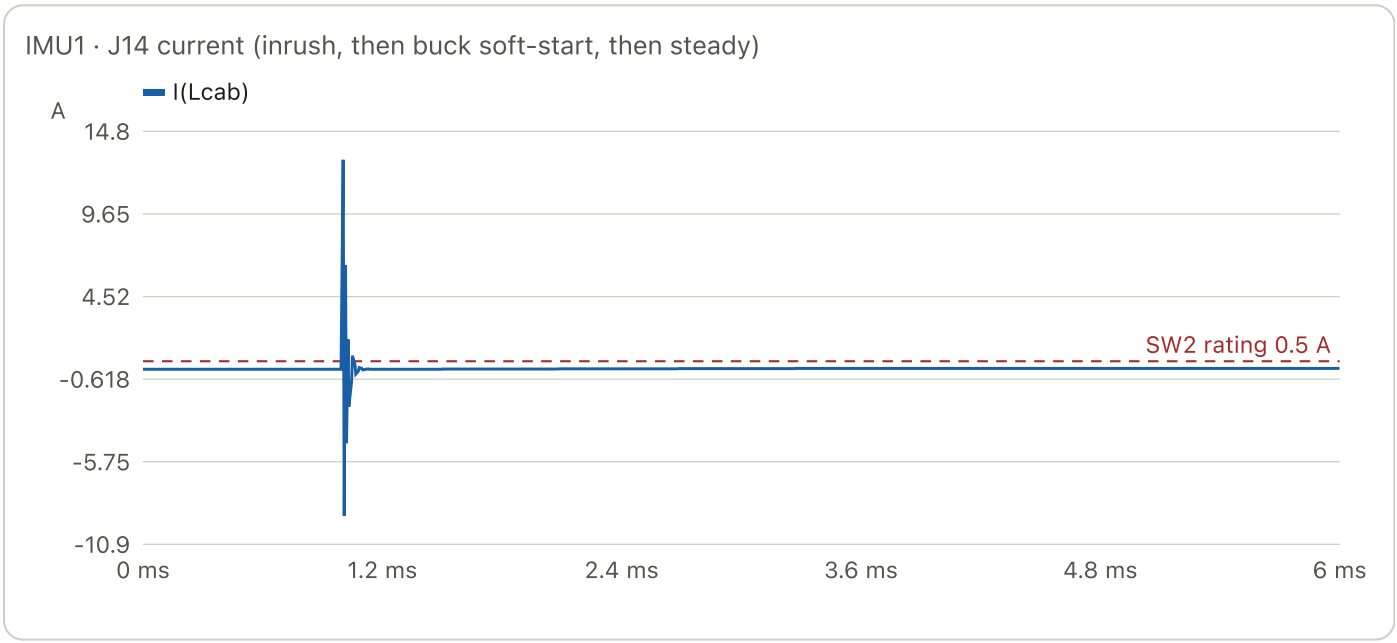
Parts: U9 URB2409S-10WR3 9 V → C29 → SW2 SS12D10G5 → LED8 / R35 → J14 → 1 m cable → **C-Series Connectivity Board** (part of the 6212-3009 USB development kit) Vaux input → its TPS54260 buck → 5 V → **3DM-CV7-AHRS** (3.2–5.2 V, 0.23 W, per HBK's product page and the board's Interface Control Document).

IMU1 · switching the IMU on (`imu1_devkit_switch_on.net`)

SW2 closes with U9's 9 V already up. The 1 m cable and the board's ceramic input capacitors (2 × 2.2 µF + 0.1 µF) ring: Vaux overshoots to 14.3 V, inside the board's 5.2–28 V Vaux range. The inrush through SW2 peaks at 9.1 A for a few microseconds (0.3 mJ of capacitor charge). Then the buck soft-starts, and the steady draw is 44

mA at J14, 48 mA through SW2 with LED8. That's about a tenth of SW2's 0.5 A rating, so review item L8's SW2 check is closed. U9 (1.11 A) runs at about 4 % load.





Wiring rules from the dev kit's Interface Control Document, not things a simulation can show:

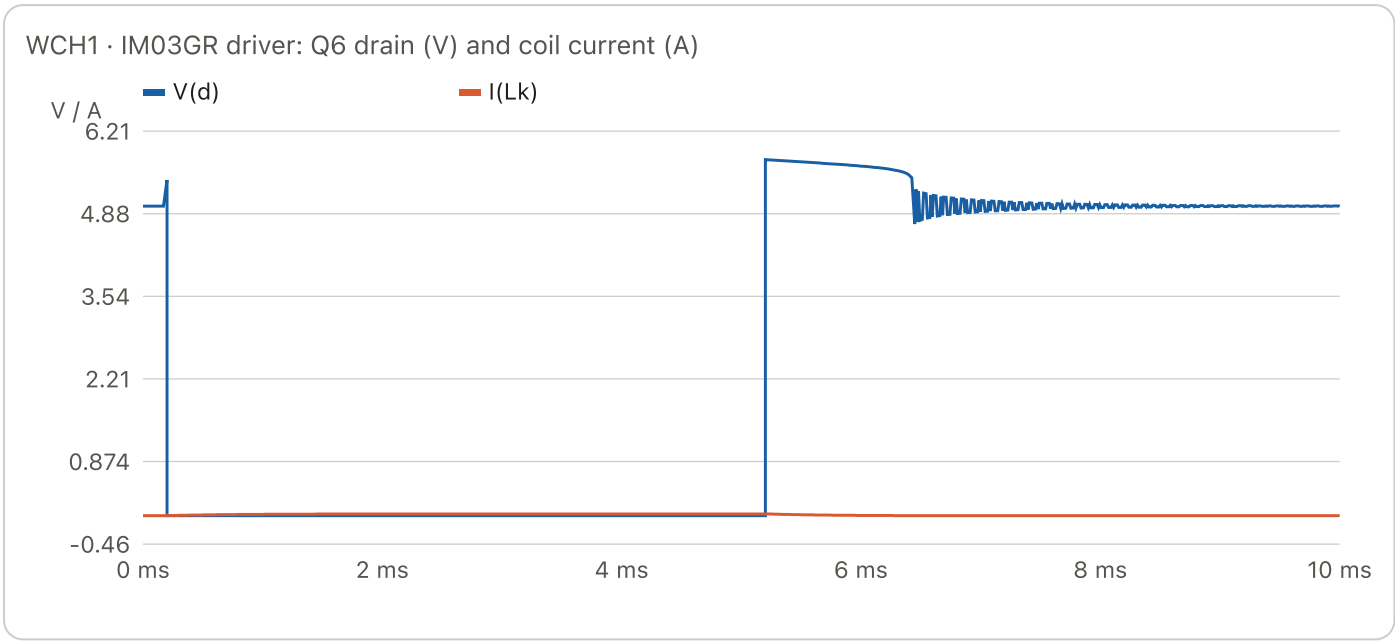
- **Feed J14's 9 V into the board's Vaux input** (micro-DB9 pin 6, "+4 V to +35 VDC" in the pin table, 5.2–28 V in the schematic notes). **Never into Vpri** (pin 3, 3.2–5.2 V): Vpri goes straight to the CV7, and 9 V there would destroy it.
- **Vaux or Vpri, never both.** The USB dev cable (micro-DB9 to USB) powers Vpri from the PC's 5 V. When the USB cable is plugged in for configuration, switch SW2 off.
- J14 pin 2 is the carrier's logic GND. Over USB the IMU also connects to the PC's ground: normal, but keep the IMU's USB and power cables together.

11 · Winch · sheet Winch (page 10)

Parts: PA0 → R43 100 Ω (R44 100k) → Q6 AO3400A → K5 IM03GR coil (178 Ω), D25 1N4148W flyback. K6 / PA1 is identical. The contact side (J18) is isolated and passive.

WCH1 · coil driver (`wch1_coil_driver.net`)

The coil draws 28 mA. At turn-off, D25 clamps the drain at 5.75 V, and the current falls to 3 mA in 0.87 ms, so release is quick.

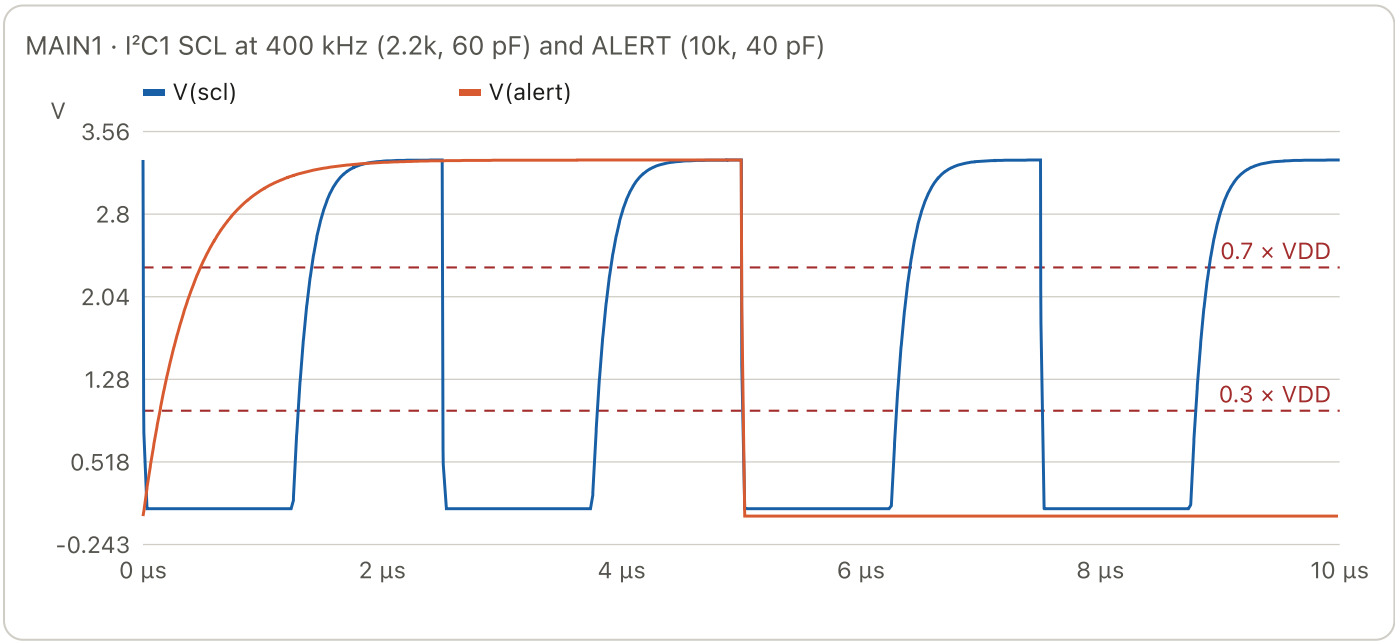


12 · Main · sheet Main (page 2), devices on Relays (pages 5–6)

Parts: R3/R4 2.2k I²C1 pull-ups and R5 10k ALERT pull-up to +3V3_MOD; four INA236 (U2–U5) and the STM32 on the bus.

MAIN1 · I²C and ALERT (`main1_i2c.net`)

400 kHz with an estimated 60 pF of bus capacitance. SCL rises (30–70 %) in 112 ns, against the 300 ns fast-mode limit, and the LOW level is 0.09 V. ALERT (10k, 40 pF) rises in 0.34 µs, which is plenty for an interrupt or polled line.



13 · Not simulated, and why

Sheet (page)	Circuit	Reason
CAN (3)	DB9 + RJ45, H24VNT2B ESD	Passive on the carrier: the transceiver and terminator are on the MES4-MCU module, which has its own review
AuxPower (9)	USB-C CC (R40/R41), shield R42/C37	Static: 10k Rp to 5 V advertises 3 A; nothing changes over time
LoadCell (8)	ADS1232 digital interface (PA4–PA6)	Digital at 10 SPS, handled in firmware
Winch (10)	Contact interlock, J18	Logic of the contacts, traced in the design review; isolated 48 V side
Main (2)	ST-Link, reset, ESD	Passive

14 · Models and assumptions

- Values come from the schematic netlist. Ceramic 10 µF / 50 V parts are modelled at ≈ 4 µF at 24–30 V, and 22 µF / 25 V at ≈ 15–18 µF at 5 V (DC bias).
- Electrolytic ESR: 100 µF / 63 V 0.25 Ω, 47 µF / 100 V 0.6 Ω (datasheet-class, not measured).
- Supply: DDR-480C-24 at 24 V (10 mΩ); pack 29.4 V full (7S Li-ion; 8S LiFePO4 would be 29.2 V), 60 mΩ assumed; XY-CD63L switch 5 mΩ assumed; AB20 Schottky diodes ≈ 0.45 V at 10 A (assumed; Autoro gives no drop figure). Cable from the AB20 to the carrier 2 m (2 µH, 33 mΩ).
- SMBJ30A: Zener model fitted to Vc 48.4 V at 12.4 A. AO3400A: generic VDMOS (Vth 1.05 V). 1N4148W and SS34: generic diode models. PC817: behavioural, CTR 50 %, 20 µs pole.
- Converters are constant-power loads above 9 V (U1 3.75 W, U9 0.5 W for the CV7 dev kit, U10 17 W), with 5 ms soft start (U10's output ramp 2 ms in AUX1). Screen input 100 µF, drawing 3 A.
- Assumed inductances (not on the datasheets): SRD coil 0.2 H, IM03GR 0.1 H, battery-stopper solenoid 50 mH. Inductors are linear (no saturation).
- Bus capacitance: I²C 60 pF, ALERT 40 pF, PA3 7 pF. UART cable 0.3 m (100 Ω, 1.5 ns).

15 · Schematic

The updated schematic (11 A3 pages, KiCad 9.0.2, ERC 0 errors / 0 warnings, 198 parts) follows on the next pages.

Rerun: `python3 run_sims.py && python3 make_report.py` in `sim/`. Open any `.net` file in LTspice to see every waveform; `results.txt` has the raw measurements.

How the MES4-Carrier-EC works

24 V 5 V signals and buses

+24V rail: relay contacts, INA236 monitors, sensor supplies, servo and aux power

24 V DC in (J5, XT30)

XT30 → F1 15A fuse → Q1 reverse-polarity P-FET → D9 SMBJ30A TVS + C1 100 µF bulk → +24V rail.
LED3 = 24 V present.

+24V

Filter + 24 V → 5 V (U1)

L1 pi filter → URB2405S-6WR3 (isolated, 1.2 A) → +5V_24: relay coils and the module feed.

+5V_24

Module 5 V feed

F2 1.1 A PTC + FB1 ferrite pi → +5V: module fingers 20 / 22 and the ADS1232 excitation.

+5V

ST-Link + reset (J2, SW1)

STDC14: SWD, NRST, VCP.
SW1 pulls NRST low.
H5VSD3B ESD on every pin.

SWD, NRST, VCP

CAN (J3 DB9, J4 RJ45 x4)

DB9 male to the IPC (1 GND, 3 H, 5 L) + 1x4 RJ45 (1 H, 2 L, 3 GND), one bus.
H24VNT2B ESD at each, no series R.

CAN

MES4-MCU module (J1, M.2 socket)

STM32G0B1, 64 MHz, CAN transceiver.
Gets +5V on fingers 20 / 22.
Makes its own 3.3 V → +3V3_MOD (finger 4, ≤ 50 mA) for the I²C pull-ups, INA236s, PC817 outputs and the ADS1232 logic.

All 14 GPIOs are used (GPIO map on the Main sheet).

LED1 = module +5V, LED2 = 3V3.
Hold-down: H1 M3 standoff.

No hot-plug: unplug the XT30 before fitting a module.

PA8 / PC6 / PB3 / PA7

I²C1 + PA15

PB6 / PC15

PA0 / PA1

PA2 / PA3

PA4-PA6

4 relay outputs (CIF §7)

1 safety light: PA8 drive, INA236 0x40
2 LED DRL: PC6 drive, INA236 0x41
3 battery stopper: PB3 drive, INA236 0x42
4 spare: PA7 drive, INA236 0x43

GPIO → 100 Ω → AO3400A → 5 V coil (SRD-5VDC-SL-A, on +5V_24).
+24V → 30 mΩ shunt → contact → 3A PTC → output (2 A continuous).
No coil feedback (pins used elsewhere).
INA236 per channel: load current + 24 V rail, ALERT → PA15.

2 Autonics sensor inputs

+24V → 100 mA PTC → sensor supply.
Sensor OUT → 10k → PC817 → PB6 / PC15 (HIGH = sensor ON). PNP 24 V sensors.

Load cell (U8 ADS1232)

4-wire bridge, excited from +5V (ratiometric, as on the MES4-Flasher).
100 Ω / 100 nF / 10 nF filters, BAV199 clamps. 24 bit, gain 128, 10 SPS.
DRDY PA4, SCLK PA5, PDWN PA6.

Winch control (page 10)

Two IM03GR relays, as on the MES2_Extension:
PA0 → CLK, PA1 → ACLK (AO3400A drivers).
Contacts switch the 48 V winch controller's direction inputs (J18).
Hardware interlock: both on = both open.

Door servo (page 9)

J15: +24V → F9 3 A PTC → FE-URT-2 → Feetech servos (24 V types).
J17: USART2 PA2 TX / PA3 RX (100 Ω / 1k) + ESD.
FE-URT-2 set to 3.3 V levels; its 5 V logic supply comes from the screen converter.

IMU 9 V + screen 5 V (page 9)

Own L pi filter from +24V.
U9 URB2409S-10WR3 → 9 V → SW2 switch → J14 IMU terminal (as on the MES3.1 carrier).
U10 URB2405YMD-20WR3 → 5 V 4 A → 3 A PTC → J16 USB-C (Rp 10k: 5 V / 3 A) → ARZOPA Z1C screen.

Notes

E-Crawler carrier for one MES4-MCU module (MES4-CIF rev A-d6). Circuits taken from the MES4-Flasher: PC817 sensor inputs, ADS1232 load cell, relay channels (driver, current monitor) and the XT30 / URB2405S 24 V input.
GPIO budget: 4 relay drives + ALERT + 3 ADS1232 + 2 sensors + winch (2) + servo USART2 (2) = 14.
Differs from the MES4-CIF §2c table: no relay coil feedback, so PA0-PA3 drive the winch and the servo UART.
Relay drive per CIF §7a (AO3400A, 100 Ω, 100k pull-down); the Flasher uses a 2SC2884 BJT for the same job.
No hot-plug: unplug the 24 V XT30 before inserting or removing the module.
Pages: Main (2), CAN (3), Power (4), Relays 1-2 (5), Relays 3-4 (6), Sensors (7), LoadCell (8), AuxPower (9), Winch (10), Pinout (11).

Pages

Main	CAN	Power	Relays1-2	Relays3-4
File: Main.kicad_sch	File: CAN.kicad_sch	File: Power.kicad_sch	File: Relays12.kicad_sch	File: Relays34.kicad_sch

Sensors	LoadCell	AuxPower	Winch	Pinout
File: Sensors.kicad_sch	File: LoadCell.kicad_sch	File: AuxPower.kicad_sch	File: Winch.kicad_sch	File: Pinout.kicad_sch

E-Crawler carrier for the MES4-MCU M.2 module
Aaron Raj Bhatti
Monkey Robotics
Sheet: /
File: MES4_Carrier_EC.kicad_sch

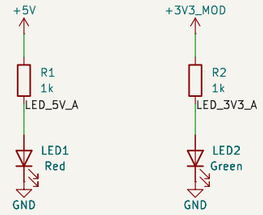
Title: MES4-Carrier-EC

Size: A3	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 9.0.2		Id: 1/11

Module GPIO map: all 14 used

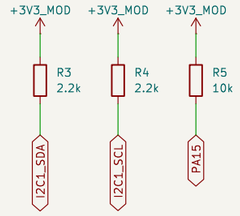
PA8 finger 31 out relay 1 drive, safety light (Q2)
PC6 finger 29 out relay 2 drive, LED DRL (Q3)
PB3 finger 61 out relay 3 drive, battery stopper (Q4)
PA7 finger 35 out relay 4 drive, spare (Q5)
PA0 finger 49 out winch CLK relay (K5)
PA1 finger 47 out winch ACLK relay (K6)
PA2 finger 45 out door servo UART TX, USART2 (J17 pin 4)
PA3 finger 43 in   door servo UART RX, USART2 (J17 pin 3)
PA15 finger 63 in   INA236 ALERT, all four wire-OR (R5 10k pull-up)
PA4 finger 41 in   ADS1232 DRDY/DOUT (U8)
PA5 finger 39 out ADS1232 SCLK (100k pull-down)
PA6 finger 37 out ADS1232 PDWN (100k pull-down)
PB6 finger 59 in   Autonics sensor 1 (U6 PC817)
PC15 finger 51 in   Autonics sensor 2 (U7 PC817)
I²C1 SDA / SCL fingers 53 / 55: INA236 0x40 / 0x41 / 0x42 / 0x43
Not used: USB 23 / 25 (open, CIF rule 8). VCP TX / RX: STDC14 only.
Differs from MES4-CIF §2c: no relay coil feedback (PA0–PA3), so those four pins drive the winch and the servo UART. Relay diagnosis uses the INA236s only.

Power indicators



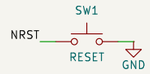
LED1 red = module +5V present (3 mA).
LED2 green = module 3V3 present
(fed from finger 4, ~0.7 mA).

I²C1 + ALERT pull-ups



CIF rule 7: always fitted,
to the module's 3V3 (finger 4).
2.2k: 4 × INA236 at 400 kHz.
R5: INA236 ALERT wire-OR (PA15).

Reset



Pulls NRST low only.
No capacitor: C6 100 nF
is on the module (CIF §4a).

CAN bus (page 3)

Module CANH / CANL (fingers 1 / 3) go to the CAN sheet
as global labels:

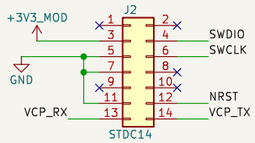
J3 DB9 male to the IPC: 1 GND, 3 CAN_H, 5 CAN_L.
J4 1x4 RJ45, no magnetics: 1 CAN_H, 2 CAN_L,
3 GND on every port (T568B: white-orange, orange,
white-green).

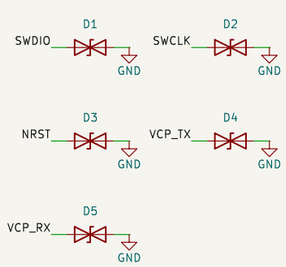
All five ports are one bus, wired in parallel.
ESD at each connector: D6, D7 (H24VNT2B).
Transceiver + 120 Ω terminator (JP1) are on the module.
Silkscreen + panel at J4: "CAN – NOT ETHERNET" (review L5).

Links to the other pages

Page 1: block diagram and page list.
Page 3 CAN: DB9 male to the IPC + 1x4 RJ45, ESD.
Page 4 Power: XT30 24 V input, fuse, reverse-polarity FET, TVS, noise filter, 24 V → 5 V.
Pages 5–6 Relays: PA8 / PC6 / PB3 / PA7 drive, 4 × INA236 (no coil feedback).
Page 7 Sensors: PB6 / PC15 (two Autonics PNP sensors).
Page 8 LoadCell: PA4 DRDY, PA5 SCLK, PA6 PDWN.
Page 9 AuxPower: IMU 9 V (MES3.1), screen USB-C 5 V / 3 A, door servo (FE-URT-2, USART2 PA2 / PA3).
Page 10 Winch: CLK / ACLK direction relays on PA0 / PA1, with a hardware interlock.
Page 11 Pinout: every module finger, its STM32 pin and what this carrier connects to it.
Signals shared between pages are global labels.

ST-Link V3MINIE (STDC14) + ESD

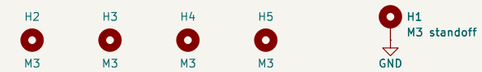




STDC14 per ST UM2448 Table 6: 3 T_VCC, 4 SWDIO, 6 SWCLK, 12 NRST,
13 T_VCP_RX (probe → PA10), 14 T_VCP_TX (probe ← PA9), 5/7/11 GND.
Never tie SWD, NRST or VCP to GND (MES3 bug). No BOOT0 button (CIF §4).

D1–D5: H5VSD3B ESD (5 V, 1.5 pF), pin 2 to GND.

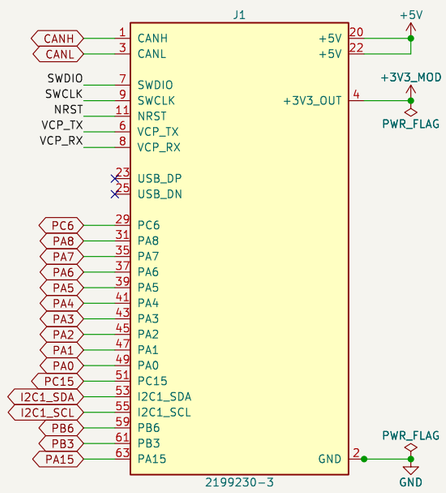
Mechanical



Design notes

1. Purpose: E-Crawler carrier for one MES4-MCU module (follows MES4-CIF rev A–d6). Circuits from the MES4-Flasher: relays, opto inputs, load cell, 24 V in.
2. No hot-plug (rule 5): unplug the 24 V XT30 before inserting or removing a module (no power switch).
3. +5V (fingers 20 / 22) = U1 URB2405S → F2 PTC → FB1 ferrite: 4.75–5.5 V, ≥ 300 mA (rule 2, 4).
4. +3V3_MOD = module LDO output on finger 4, never driven (rule 3, ≤ 50 mA): LED 0.7 + pull-ups ≤ 3.3 + 2 × PC817 0.7 + 4 × INA236 1.7 + ADS1232 0.1 + ST-Link VTref ≈ 10 mA.
5. NRST: SW1 and STDC14 pin 12 only pull low (wired-OR). No extra capacitance (C6 on the module).
6. ESD (rule 8): H5VSD3B on every STDC14 signal, H24VNT2B on CANH / CANL, no series R on CAN.
7. No back-powering (rule 6): carrier logic runs from +3V3_MOD (pull-ups, PC817s, INA236, ADS1232 DVDD).
8. All relay drives OFF in reset: 100k gate pull-downs.
9. Parts: LCSC numbers in each symbol's LCSC field.

MES4-MCU module socket (TE 2199230-3, M.2 B-key 4.2 mm)



Odd fingers (card top) on the left, power on the right.
+5V = fingers 20 + 22. GND = all other fingers + the hold-down pads (stacked pin). USB 23/25 open (CIF rule 8).
Module hold-down: H1 M3 × 2.5 mm SMD standoff + M3 screw, 40.25 mm from the socket posts (42 mm card).

Board connectors

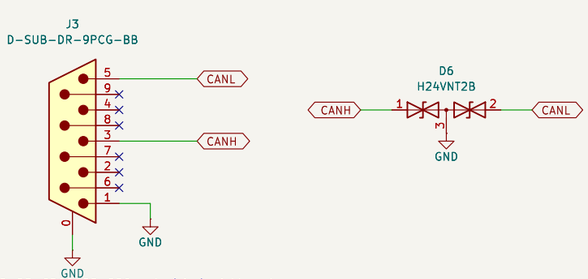
J5 24V IN, XT30 male right-angle: 1 = +24 V (flat end), 2 = GND_24V (chamfered end).
J6 / J7 / J8 / J9 RLY1–4 OUT (KF250NH 2-pin): 1 = switched +24 V, 2 = GND_24V.
J10 / J11 SENS1 / SENS2 (KF250NH 3-pin): 1 = +24 V (100 mA PTC, brown), 2 = output (black), 3 = 0 V (blue).
J12 LC SIG: 1 SIG+ green, 2 SIG− white, 3 shield.
J13 LC EXC: 1 EXC+ red, 2 EXC− black, 3 shield. 5 V load cell only: never 24 V.
J3 CAN DB9 male (IPC): 1 GND, 3 CAN_H, 5 CAN_L.
J4 CAN RJ45 x4: 1 CAN_H, 2 CAN_L, 3 GND (T568B).
J14 IMU 9V (KF250NH 2-pin): 1 = +9 V after the switch, 2 = GND. To the CV7 dev board Vaux only.
J16 SCREEN USB-C: 5 V / 3 A power only (no data).
J15 SERVO 24V: 1 +24 V (3 A PTC), 2 GND_24V.
J17 SERVO UART: 1 GND, 2 +5 V, 3 RX, 4 TX.
J18 WINCH: 1 CLK, 2 COM, 3 ACLK (relay contacts).
J2 STDC14: ST-Link V3MINIE (SWD, NRST, VCP).
Test points: Power sheet.

Sheet 2: module socket, ST-Link, reset, I²C pull-ups, power LEDs
Aaron Raj Bhatti
Monkey Robotics
Sheet: /Main/
File: Main.kicad_sch

Title: MES4-Carrier-EC

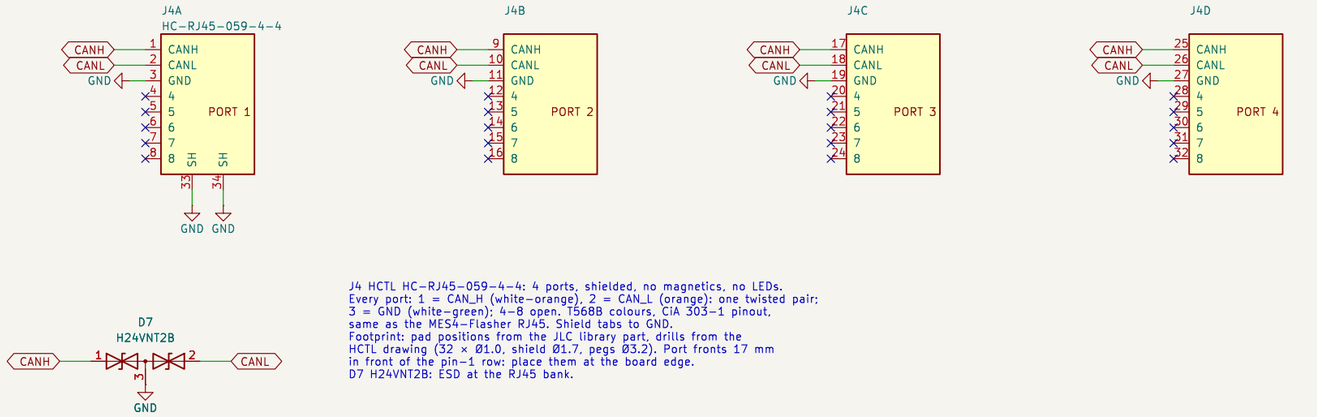
Size: A3	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 9.0.2		Id: 2/11

IPC link: DB9 male



J3 XUNPU D-SUB-DR-9PCG-BB: DB9 male (pins), right angle.
4-40 shell, KiCad DSUB-9 footprint (EdgePinOffset 7.70 mm).
Pinout per the IPC: 1 = GND, 3 = CAN_H, 5 = CAN_L, the rest open.
Shell / mounting holes to GND (as the RJ45 shield).
Male on the board and on the IPC: use a straight-through
female-female DB9 cable (1-1, 3-3, 5-5).
Not the CiA 303-1 DB9 pinout (2 CAN_L, 7 CAN_H, 3 GND):
a standard CAN DB9 cable or adapter will not match.
D6 H24VNT2B: ESD at the connector, no series R (CIF rule 8).

CAN ports: 1x4 RJ45, no magnetics



J4 HCTL HC-RJ45-059-4-4: 4 ports, shielded, no magnetics, no LEDs.
Every port: 1 = CAN_H (white-orange), 2 = CAN_L (orange): one twisted pair;
3 = GND (white-green); 4-8 open. T568B colours, CiA 303-1 pinout,
same as the MES4-Flasher RJ45. Shield tabs to GND.
Footprint: pad positions from the JLC library part, drills from the
HCTL drawing (32 × Ø1.0, shield Ø1.7, pegs Ø3.2). Port fronts 17 mm
in front of the pin-1 row: place them at the board edge.
D7 H24VNT2B: ESD at the RJ45 bank.

CAN bus rules (CIF §5, rules 8-9)

All five ports (DB9 + 4 × RJ45) are the same bus, wired in parallel to the module's CANH / CANL (fingers 1 / 3).
Keep the bus a line: IPC → DB9 → this board → one RJ45 → next node → ... A port used as a branch is a stub:
keep each stub ≤ 0.3 m at 500 kbit/s. More long branches need a hub with its own transceivers, not more ports.
Exactly two 120 Ω terminators, at the two physical ends of the bus: bridge JP1 on the module only if this board is an end.
All nodes on one solid ground: pin 3 (RJ45) / pin 1 (DB9) carry it. Check with power off: CAN_H-CAN_L ≈ 60 Ω.
Never carry 24 V in a CAN cable or connector (CIF rule 9): these jacks carry only CAN_H, CAN_L and GND.
No series resistors on CAN_H / CAN_L (CIF rule 8): the module must drive the 60 Ω bus directly.

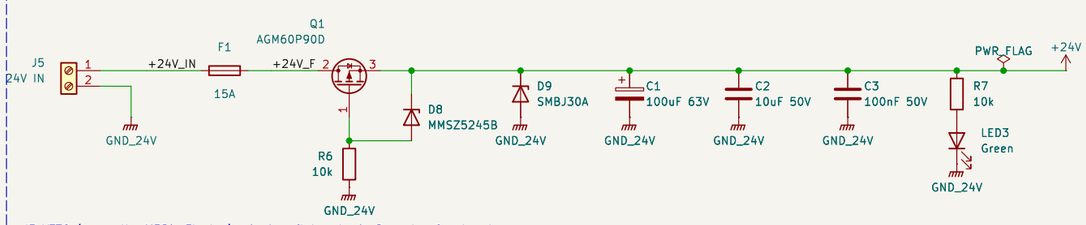
Sheet 3: CAN bus connectors: DB9 male to the IPC + 1x4 RJ45 (no magnetics), ESD
Aaron Raj Bhatti
Monkey Robotics

Sheet: /CAN/
File: CAN.kicad_sch

Title: MES4-Carrier-EC

Size: A3	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 9.0.2		Id: 3/11

24 V input: XT30, fuse, reverse-polarity FET, TVS, bulk



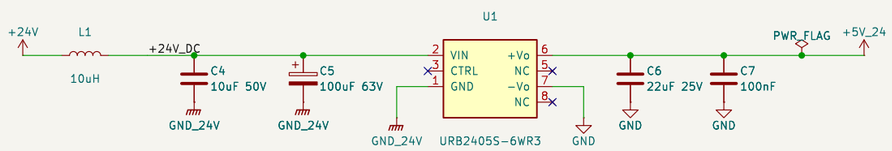
J5 XT30 (as on the MES4-Flasher): pin 1 + flat end, pin 2 − chamfered end.
F1 15A fast fuse (Nano2 2410, 65 V): protects the XT30 and the board copper; a realistic 6–8 A
stays under its 75 % derating (11 A); all loads at their limit (≈ 13 A, see the budget) would not. Each relay output has its own 3 A PTC, so a load short never blows F1.
Q1 P-MOSFET reverse-polarity block: correct polarity turns it on (Vgs clamped by D8,
R6 10k to GND_24V); a reversed lead leaves it off. It replaces the Flasher's SS56 Schottky,
which would dissipate ≈ 3–4 W here. AGM60P90D: −60 V, 13 mΩ: ≈ 0.8 W at 8 A, ≈ 2.2 W at 13 A: size the drain-tab copper for 2.2 W.
D9 SMBJ30A (600 W): surges and spikes from motors and solenoids on the 24 V bus (CIF §7a).
C1 100 µF / 63 V electrolytic: bulk, and its ESR damps the cable inductance (hot-plug ringing).
C2 / C3: HF bypass. LED3 green = 24 V present (after Q1).
+24V feeds the relay contacts, the INA236 current monitors and the sensor supplies.

Power budget and noise strategy

24 V current: relays ≤ 4 × 2 A (CIF §7; realistic 3–5 A: lamp ~0.3 A, DRL, 1.5 A solenoid), sensors
2 × ≤ 100 mA, logic DC/DC ≈ 0.15 A, IMU 9 V ≈ 0.02 A (CV7 0.23 W) + screen 5 V ≤ 0.95 A + door servo ≤ 3 A (page 9).
Realistic total ≈ 6–8 A. Every load at its limit at once (≈ 13 A) would exceed F1 15A's 75 % derating
(11 A) and the XT30's 15 A rating: don't stall the servo while all four relays carry 2 A.

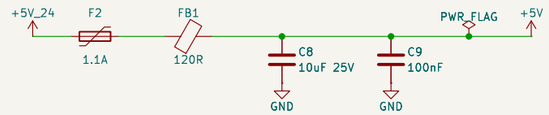
Noise and transients, from the cable inwards:
1. D9 SMBJ30A clamps surges and inductive spikes on the 24 V bus.
2. C1 electrolytic damps the cable LC: no ringing above 24 V on hot-plug.
3. C2 / C3 ceramics: low impedance for HF at the input.
4. Pi filter C2 + L1 + C4 / C5 feeds only the DC/DC: the logic supply is decoupled
   from the bus, and the converter's switching noise stays off it.
5. Isolated DC/DC: the grounds meet only at NT1, so relay load currents never flow in GND.
6. Pi filter F2 + FB1 + C8 / C9 on the module 5 V, and FB2 again on the ADS1232 AVDD.
7. Flyback diodes at every relay coil (1N4148W) and every output (SS34).
8. Sensor supplies: 100 mA PTC + 100 nF / 50 V at each terminal.

Noise filter + isolated 24 V → 5 V



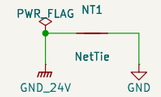
L1 10 µH + C4 / C5, with C2 on +24V: pi filter in front of the DC/DC only.
Relay contacts, INA236s and sensors use the unfiltered +24V.
U1 URB2405S-6WR3 (as on the MES4-Flasher): 9–36 V in, 5 V 1.2 A, short-circuit protected.
CTRL open = on. C5 = the maker's 100 µF input cap; C6 / C7 = output caps.
The maker's EMC circuit adds a 470 µH common-mode choke + 1 nF Y caps: not fitted, because
NT1 ties the grounds (the 1.5 kV isolation is not used). The maker lists hot-plug as unsupported:
L1 slows the inrush into C5; Q1 / D9 / C1 take the XT30 plug-in.
Load: module ≤ 240 mA + 4 coils 285 mA + 2 winch coils 56 mA + load cell 13 mA ≈ 0.6 A (50 %).

Module + ADC 5 V feed (CIF rule 4)



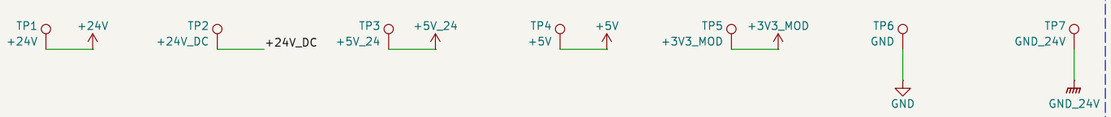
F2 1.1 A PTC: the fuse per module slot (CIF rule 4).
FB1 + C8 / C9: pi filter, so relay-coil switching on
+5V_24 stays off the module and the load-cell excitation.
+5V → module fingers 20 / 22 (4.75–5.5 V, CIF rule 2) and the
ADS1232 AVDD (LoadCell sheet). Place C8 / C9 at the socket.

Grounds



GND_24V: 24 V return (XT30 pin 2, relay outputs,
sensor 0 V, TVS, bulk caps, DC/DC input side).
GND: logic (module, INA236, ADS1232, DC/DC output).
NT1: the only joint between them (the INA236s
measure +24V against GND). Place it at the DC/DC.

Test points (bring-up)



2 mm pads with a 1 mm hole: probe tip, or solder a wire loop for clips (not in the BOM).
Clip the scope ground to GND for the module side and to GND_24V for the 24 V side.

First power-up

1. No module fitted. Apply 24 V on J5 from a current-limited supply: LED3 lights;
   +24V, +24V_DC, +5V_24 and +5V present (LED1 on).
2. Reverse the lead briefly: nothing lights and +24V stays at 0 V (Q1 blocks).
3. Unplug 24 V, fit the module, plug in again: LED2 lights (module 3V3).
   Then flash over STDC14 (J2).

Sheet 4: 24 V input + noise filtering, isolated 24 V → 5 V, module 5 V feed
Aaron Raj Bhatti
Monkey Robotics
Sheet: /Power/
File: Power.kicad_sch

Title: MES4-Carrier-EC

Size: A3	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 9.0.2		Id: 4/11

Relay diagnosis (CIF §7b, INA236 only)

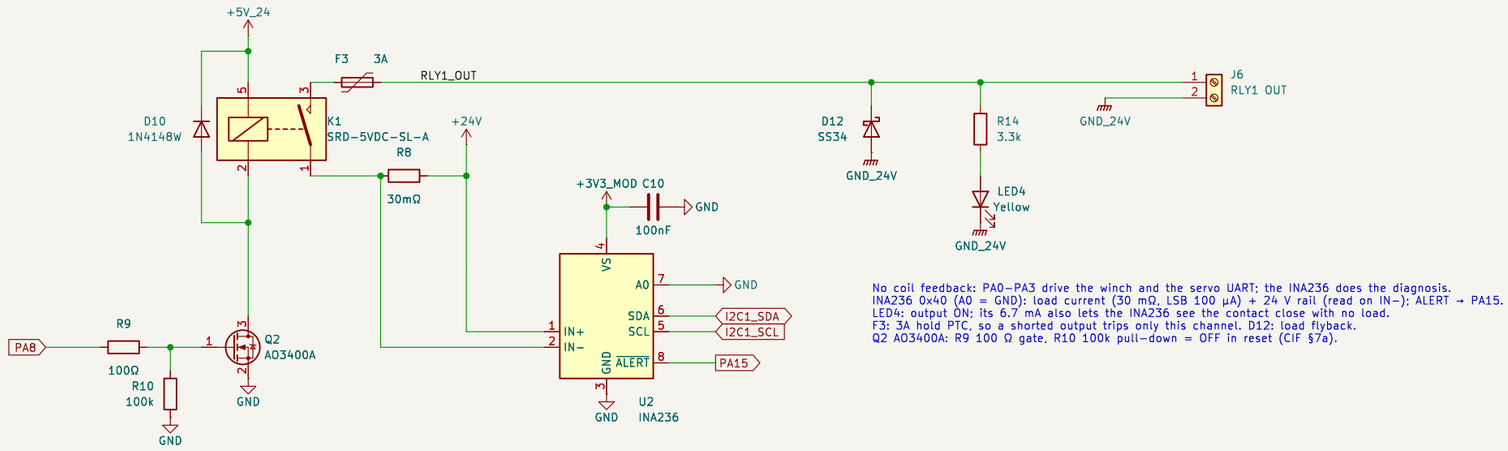
Coil supply = +5V_24 (U1 DC/DC, Power sheet).
Read ≥ 30 ms after switching.

OFF, I ≈ 0 → OK, relay off
OFF, I > 3 mA → contacts welded or driver shorted
ON, I in window → OK (LED 6.7 mA + load)
ON, I ≈ 0 → relay not closing: open coil, driver or GPIO, no +5V_24, contact, output PTC or input fuse open, or no 24 V
ON, I over limit → short / stall (ALERT → PA15)
any, VBUS < 20 V → 24 V supply fault

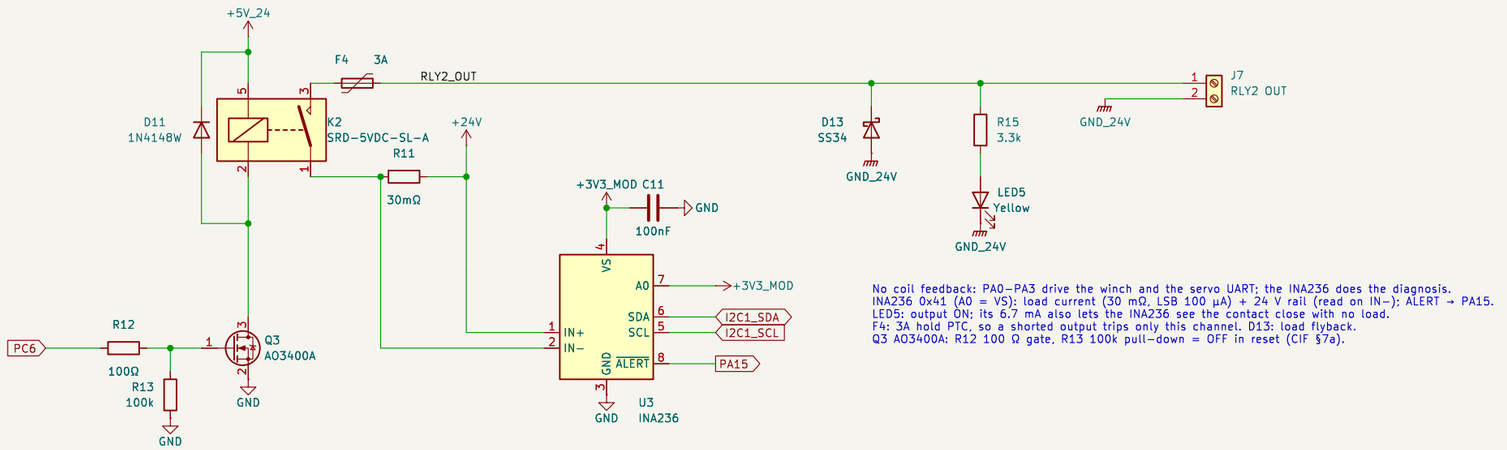
No coil feedback on this board (differs from CIF §2c / §7a): PA0 / PA1 drive the winch and PA2 / PA3 are the servo UART (USART2). So an open coil or a dead driver looks the same as a contact that does not close: "ON, I ≈ 0". I = INA236 load current, VBUS = 24 V rail.

INA236 (30 mΩ; CIF §6a names the INA226, swapped for its 50 V rating, review M1): CONFIG 0x4327, CAL 1707 (LSB 100 µA, FS 2.73 A), MASK/ENABLE 0x8001 (SOL, latched), ALERT limit 30000 = 2.5 A default; set each channel's value from the CANopen dictionary. "Current present" threshold 3 mA (the LED alone draws 5.1–6.7 mA over 21–29 V). On ALERT LOW: read MASK/ENABLE of each INA236 (AFF bit 4) to find the channel; the read also clears the latch. Switch it OFF, report on CAN. Unlike the INA226: bus LSB 1.6 mV, bus read on IN− (no VBUS pin), Device ID 0xA080.

Relay 1 (safety light): drive PA8, INA236 U2 at 0x40



Relay 2 (LED DRL): drive PC6, INA236 U3 at 0x41



Sheet 5: relay outputs 1–2 (safety light, LED DRL) with load–current feedback (INA236)
Aaron Raj Bhatti
Monkey Robotics
Sheet: /Relays1–2/
File: Relays12.kicad_sch
Title: MES4–Carrier–EC

Size: A3	Date: 2026–10–27	**Rev: A**
KiCad E.D.A. 9.0.2		Id: 5/11

Loads (CIF §7, all 24 V DC)

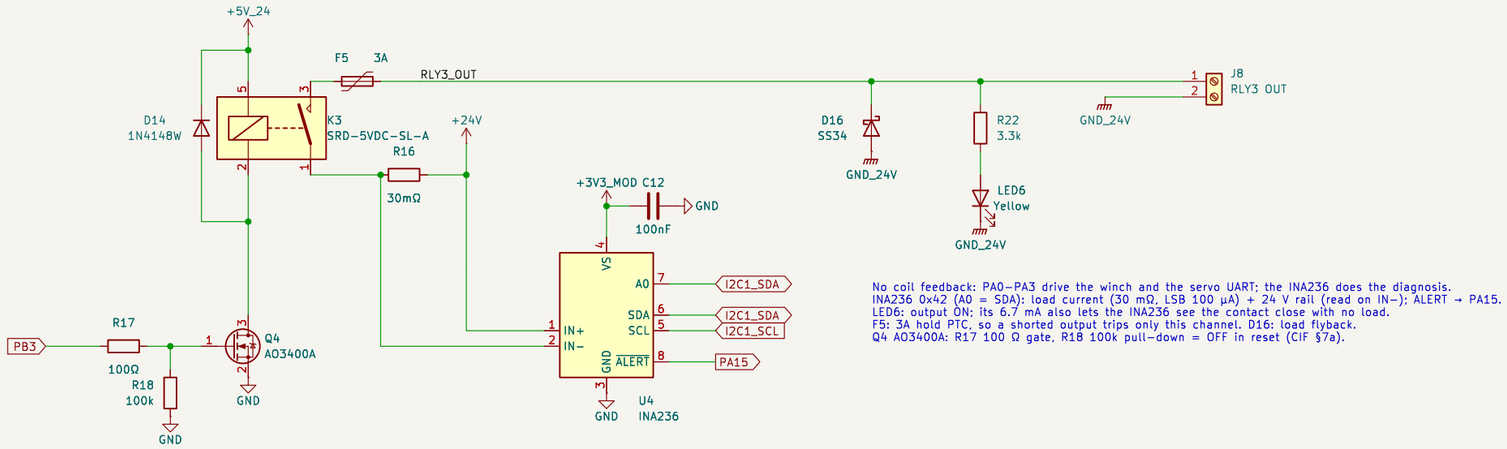
CH1 safety light: Hanyoung T084-PAD24Y-D
(LED + motor). Current TBD: measure,
then set its window (est. 0.1–0.3 A).
CH2 LED DRL: ≤ 2 A continuous.
CH3 battery stopper: Heschen HS-1564B 24 V
push-pull solenoid, 1.5 A, intermittent
(80 °C after 3 min continuous): firmware
enforces a maximum on-time + cool-down.
CH4 spare: ≤ 2 A continuous.

Every channel: 24 V / 2 A continuous,
3A hold PTC per output (CIF §7a: fuse
per channel). Shunt on the supply side of the
contact, so the INA236 never sees the load
flyback (−0.7 V < −0.3 V input limit) and a
welded contact still shows current while OFF.

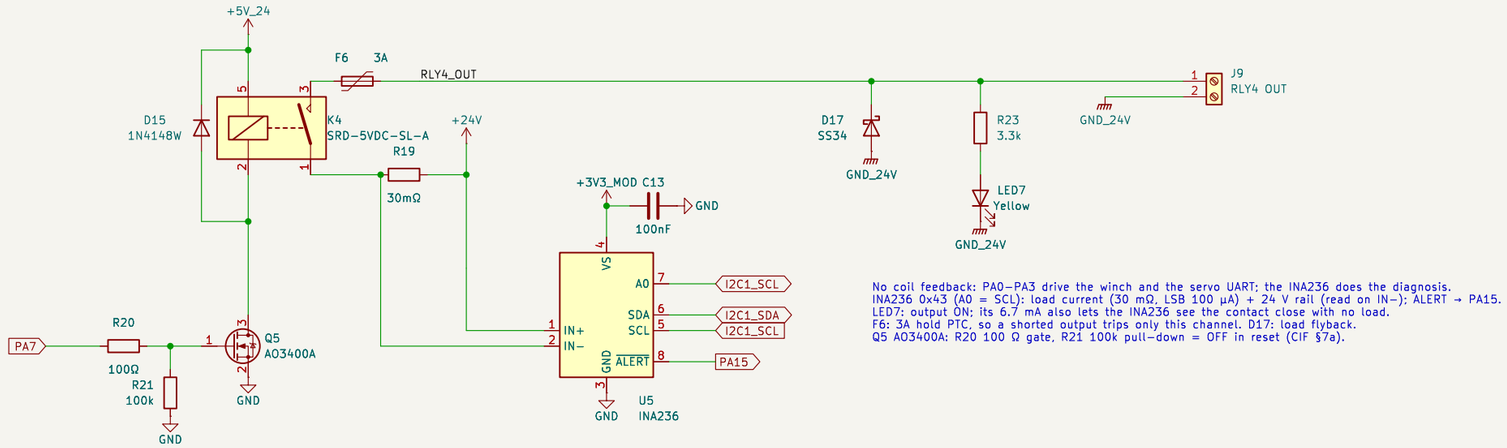
Inductive loads (solenoid): SS34 flyback at
each output on the board; also fit an SS14-class
diode at the solenoid itself.

Coil budget: 4 × 71 mA ≈ 285 mA from +5V_24
(U1 URB2405S, 1.2 A).
All relays OFF in reset / boot / firmware
update: 100k gate pull-downs; PA8 reads LOW in
reset (UCPD pull-down). PA7 is pulled DOWN by
the bootloader (CIF §2c).

Relay 3 (battery stopper): drive PB3, INA236 U4 at 0x42



Relay 4 (spare): drive PA7, INA236 U5 at 0x43

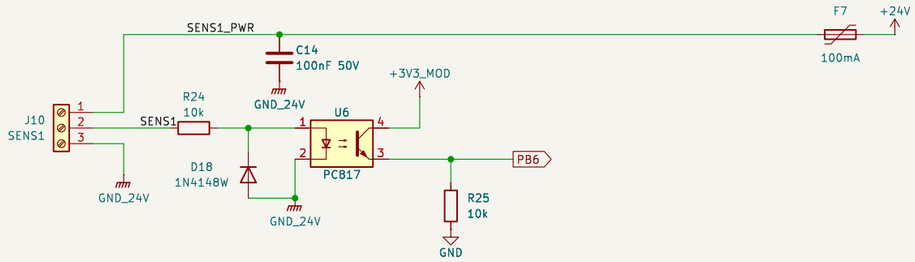


Sheet 6: relay outputs 3–4 (battery stopper, spare) with load-current feedback (INA236)
Aaron Raj Bhatti
Monkey Robotics
Sheet: /Relays3-4/
File: Relays34.kicad_sch

Title: MES4-Carrier-EC

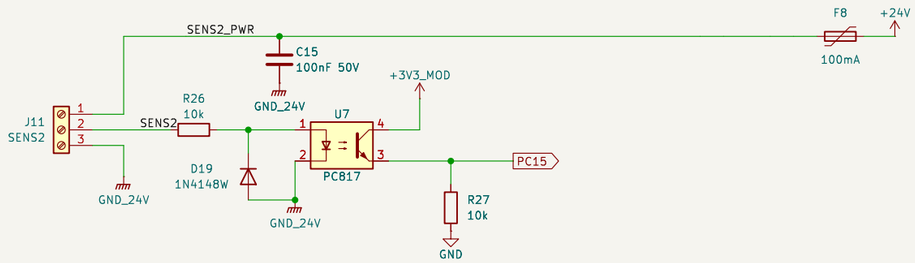
Size: A3	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 9.0.2		Id: 6/11

Sensor 1 → PB6: Autonics PNP proximity sensor, PC817 opto input



J10: 1 = +24 V (brown) through F7 100 mA PTC, 2 = sensor output (black), 3 = 0 V (blue).
Output ON (24 V): ≈ 2.3 mA through R24 10k + U6 LED → PB6 HIGH (3.3 V, emitter follower).
Output OFF, cable unplugged or broken: PB6 LOW (R25 10k pull-down).
D18 1N4148W: reverse clamp across the opto LED. C14 100 nF / 50 V: supply decoupling + ESD.
A shorted sensor cable trips F7 only: the other sensor and the board keep running.

Sensor 2 → PC15: Autonics PNP proximity sensor, PC817 opto input



J11: 1 = +24 V (brown) through F8 100 mA PTC, 2 = sensor output (black), 3 = 0 V (blue).
Output ON (24 V): ≈ 2.3 mA through R26 10k + U7 LED → PC15 HIGH (3.3 V, emitter follower).
Output OFF, cable unplugged or broken: PC15 LOW (R27 10k pull-down).
D19 1N4148W: reverse clamp across the opto LED. C15 100 nF / 50 V: supply decoupling + ESD.
A shorted sensor cable trips F8 only: the other sensor and the board keep running.

Sensor inputs: notes

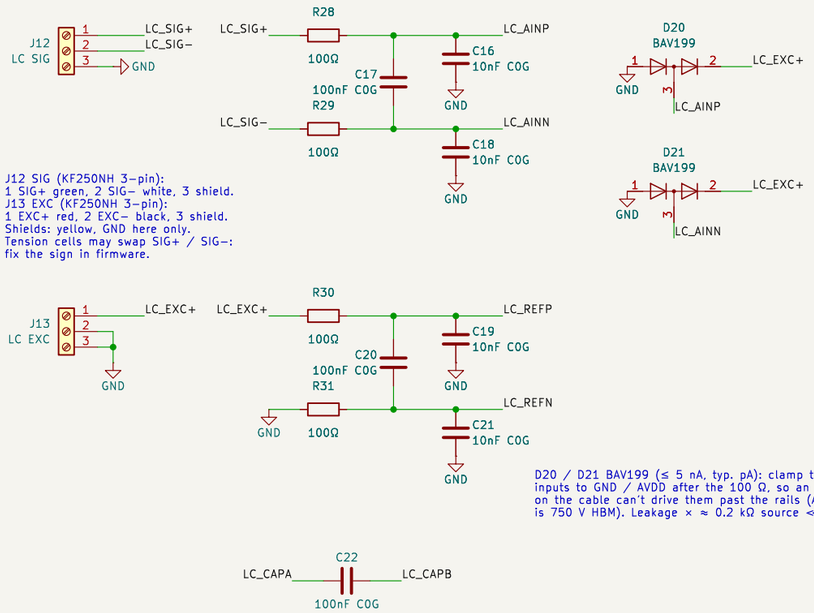
For 3-wire PNP (sourcing) 24 V sensors, e.g. Autonics PRD08 / PR series (brown +V, black OUT, blue 0 V).
NPN (sinking) sensors need a different input circuit.

Same circuit as the MES4-Flasher sensor channels 3 / 4 (PB6 / PC15), without the Flasher INA226 supply monitors.
GPIO budget: all 14 module pins are used (Main sheet, MES4-CIF §2c). More sensors need inputs on
I²C1, e.g. a TCA9535 expander (0x20-0x27), polled.

PB6: the ST bootloader enables a 25-55 kΩ pull-up on PB6 (blank chip only); with the 10k pull-down it
reads LOW, which is harmless for an input. PC15: input only (RTC-domain pin, CIF rule 10): fine here.
Interrupts: PB6 → EXTI6. PC15 shares EXTI15 with PA15 (INA236 ALERT): give EXTI15 to PC15 (sensor 2)
and poll ALERT, which the INA236 latches (MASK/ENABLE LEN = 1), so no fault is missed.
No back-powering: the PC817 collectors run from +3V3_MOD (module finger 4), so nothing drives the
module pins while it is unpowered.

Sheet 7: two Autonics PNP sensor inputs (PC817 opto, as on the MES4-Flasher)
Aaron Raj Bhatti
Monkey Robotics
Sheet: /Sensors/
File: Sensors.kicad_sch
Title: MES4-Carrier-EC
Size: A3 | Date: 2026-10-27 | **Rev: A**
KiCad E.D.A. 9.0.2 | Id: 7/11

Load cell inputs J12 / J13 + RC filters (CIF §8a)

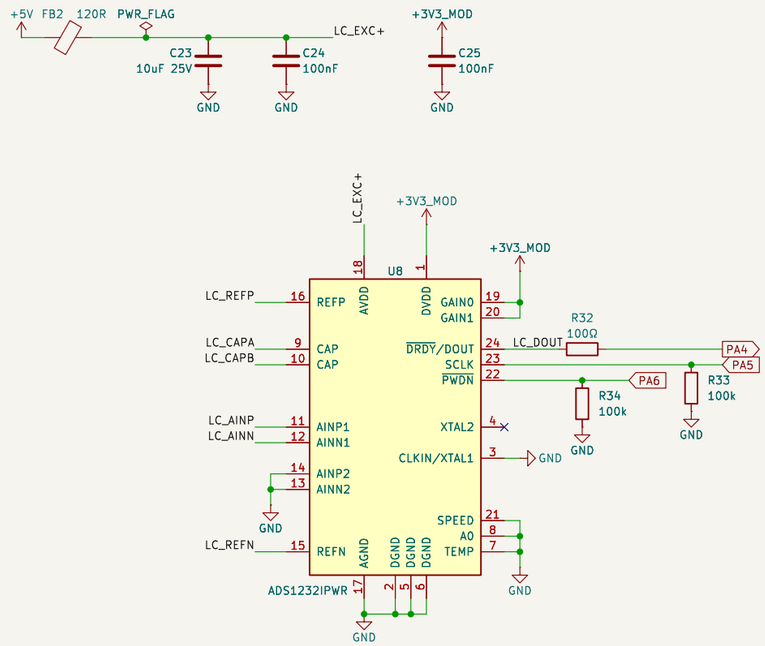


J12 SIG (KF250NH 3-pin):
1 SIG+ green, 2 SIG− white, 3 shield.
J13 EXC (KF250NH 3-pin):
1 EXC+ red, 2 EXC− black, 3 shield.
Shields: yellow, GND here only.
Tension cells may swap SIG+ / SIG−:
fix the sign in firmware.

D20 / D21 BAV199 (≤ 5 nA, typ. pA): clamp the ADC inputs to GND / AVDD after the 100 Ω, so an ESD hit on the cable can't drive them past the rails (ADS1232 is 750 V HBM). Leakage × ≈ 0.2 kΩ source ≪ 1 µV.

Signal and reference filters: 100 Ω per leg, 100 nF C0G across, 10 nF to GND (CIF §8a).
The reference gets the same network so both time constants match (TI SBAS350 §8.2).
REFP / REFN are sensed at J13 pins 1 / 2: route them as their own pair from the connector pads, not from the 5 V / GND planes. C22 = PGA bypass (pins 9–10).

U8 ADS1232 24-bit bridge ADC (TI SBAS350 weigh-scale configuration)



A0 = GND (channel 1). GAIN1 / GAIN0 = 1 / 1 → gain 128, ±19.5 mV at a 5 V reference. SPEED = GND → 10 SPS, rejects 50 and 60 Hz.
TEMP = GND. CLKIN = GND → internal oscillator, XTAL2 open. Unused channel 2 inputs to GND (TI weigh-scale example).
AVDD = +5V via FB2 = bridge excitation → ratiometric. AVDD range 2.7–5.3 V (SBAS350 §5.3).
PDWN (PA6) and SCLK (PA5) have 100k pull-downs: the ADC stays powered down until the firmware runs the power-up sequence.
DRDY/DOUT → PA4. PA4 / PA5 are 3.6 V-max pins: fine, all ADC logic runs at 3.3 V. Input common mode 2.5 V (needs 1.5–3.5 V at gain 128).

Load cell firmware (CIF §8b, TI SBAS350 §7.4.5)

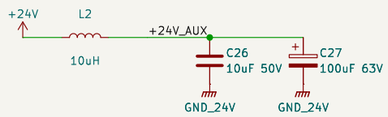
Power-up: keep PA6 (PDWN) LOW until +5V and 3V3 are stable, wait ≥ 10 µs, then PDWN HIGH ≥ 26 µs, LOW ≥ 26 µs, HIGH.
Read: wait for DRDY/DOUT LOW, clock 24 bits on SCLK (MSB first, two's complement). Clocking 26 bits instead also runs the offset self-calibration: do it after power-up and when the temperature changes. Discard the first conversions after power-up or reset.

Scale: 2 × SOP SOPLF-102 (1.5 t, 1.5 mV/V) summed in the junction box → 7.5 mV full scale at 5 V ≈ 38 % of ±19.5 mV, ≈ 2.5 µV per kg.
Bridge current ≈ 13 mA (two 780 Ω cells) from +5V. Warn at 1.5 t, alarm at 2.0 t total. Re-calibrate the scale factor (the old HX711 factor was taken at 3.3 V). Filtering as before: 0.5 kg noise threshold, ±0.25 kg stability window, 0.1 kg rounding.

5 V load cell only: J13 pin 1 is the +5V rail through FB2. Never land a 24 V wire on J12 / J13 (same push-in terminal type as the relay and sensor outputs): mark them on the silk: a filled box "5V LOAD CELL — NEVER 24V" at J12 / J13, and place them away from the 24 V terminals (review M2).
Excitation comes from +5V (Power sheet: URB2405S → PTC → ferrite pi filter), the same rail as the module.

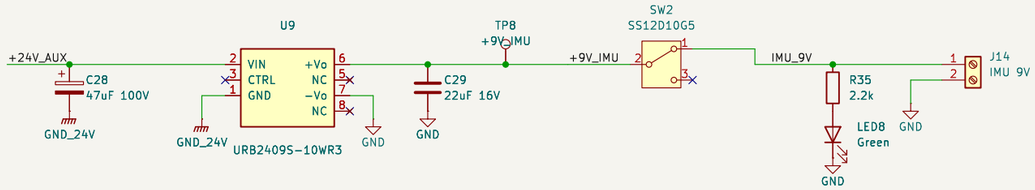
Sheet 8: load cell ADC (ADS1232, CIF §8, as on the MES4-Flasher)
Aaron Raj Bhatti
Monkey Robotics
Sheet: /LoadCell/
File: LoadCell.kicad_sch
Title: MES4-Carrier-EC
Size: A3 | Date: 2026-10-27 | **Rev: A**
KiCad E.D.A. 9.0.2 | Id: 8/11

Aux 24 V filter



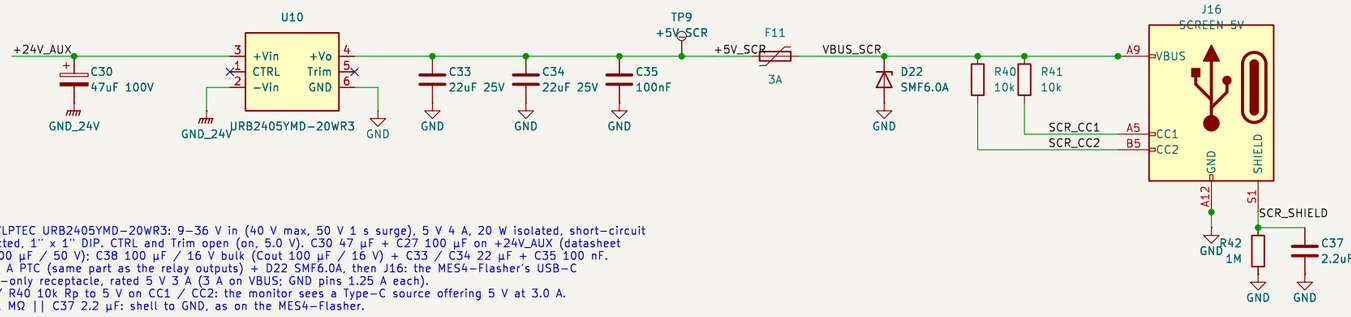
L2 10 µH (same part as L1) + C26 / C27: a pi filter
of its own for the IMU and screen converters, so the screen's
20 W converter neither loads nor disturbs the logic supply
(L1, Power sheet). Load at 24 V: IMU ≤ 0.5 A + screen ≤ 0.95 A
≈ 1.45 A; L2 is rated 2.1 A (2.9 A saturation).
Both converters are isolated: input side on GND_24V,
output side on GND (same scheme as the logic DC/DC).

IMU power: 9 V (as on the MES3.1 carrier)

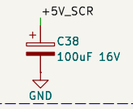


Same parts as the MES3.1−Carrier−LH "IMU Power Supply @ 9V":
U9 YLPTEC URB2409S−10WR3 (9−36 V in, 9 V 1.11 A, 10 W isolated, short−circuit protected;
MES3.1 U15). C28 47 µF / 100 V and C29 22 µF / 16 V (the datasheet's Cin / Cout).
SW2 SS12D10G5 side switch (MES3.1 U22, common = pin 2). J14 KF250NH 2−pin (MES3.1 CN10).
CTRL open = on. IMU: MicroStrain 3DM−CV7−AHRS on the C−Series Connectivity Board (6212−3009 kit), ≈ 45 mA at 9 V
(SW2 0.5 A rating OK). Wire J14 to the board's Vaux (5.2−28 V, micro−DB9 pin 6), NEVER Vpri (3.2−5.2 V, pin 3,
straight to the CV7). Vaux or Vpri, not both: switch SW2 off while the USB dev cable (5 V on Vpri) is plugged in.
LED8 green (+ R35 2.2k, 3.2 mA) sits after SW2: it lights only when the IMU terminal is powered.
LCSC stock of URB2409S−10WR3 is low (78 on 2026−10−27): order early.

Screen power: USB−C 5 V / 3 A (ARZOPA Z1C portable monitor)

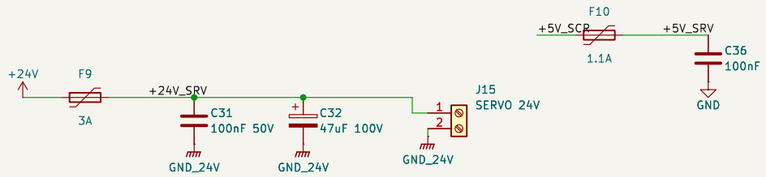


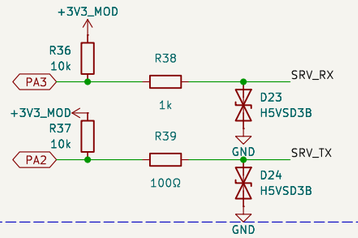
U10 YLPTEC URB2405YMD−20WR3: 9−36 V in (40 V max, 50 V 1 s surge), 5 V 4 A, 20 W isolated, short−circuit
protected, 1" x 1" DIP. CTRL and Trim open (on, 5.0 V). C30 47 µF + C27 100 µF on +24V_AUX (datasheet
Cin 100 µF / 50 V); C38 100 µF / 16 V bulk (Cout 100 µF / 16 V) + C33 / C34 22 µF + C35 100 nF.
F11 3 A PTC (same part as the relay outputs) + D22 SMF6.0A, then J16: the MES4−Flasher's USB−C
power−only receptacle, rated 5 V 3 A (3 A on VBUS; GND pins 1.25 A each).
R41 / R40 10k Rp to 5 V on CC1 / CC2: the monitor sees a Type−C source offering 5 V at 3.0 A.
R42 1 MΩ || C37 2.2 µF: shell to GND, as on the MES4−Flasher.

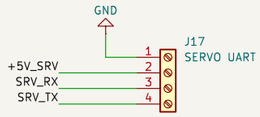


ARZOPA Z1C label: input 5 / 9 / 12 / 15 / 20 V DC, 3 A max: a USB−PD sink.
This port has no PD controller, so it offers 5 V only (15 W max).
The sibling Z1RC draws about 9 W (≈ 1.8 A at 5 V), so 5 V / 3 A should
run it; check full brightness on the bench. If it needs 9 V or more,
the port needs a USB−PD source controller instead.
VBUS is always on (no cold−socket switch): plug in the monitor, or
another sink, only; never a charger or a laptop.
Power only: the video goes over the monitor's own HDMI / USB−C
from the IPC. Rp is to 5 V: the monitor sinks at most 3 A.

Door servo: Feetech, via the FE−URT−2 board (as on the MES2 Harvester)







FE−URT−2 mount: H6−H9 M3 x 2 mm SMD standoffs (SMTSOM320BTR)
on its 48.5 x 28.7 mm holes, M3 screws from the top. No parts under it.
GPS (ArduSimple simpleRTK4 Dual) stacked above it: H10−H12 M3 x 6.5 mm
SMD standoffs (SMTSOM365BTR) + M3 x 12 mm male−female spacers (fitted by
hand) on its left−bottom, left−top and right holes; underside 18.5 mm up.
The RTK is powered from its own GPS USB−C. Standoffs not connected.

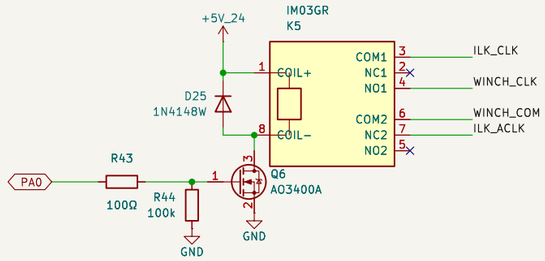


J17 SERVO UART: same pinout as CN13 on the MES2 Harvester.
J15 SERVO 24V: pin 1 = +24 V like every power terminal here
(Harvester CN5 is reversed: re−pin that cable).
The FE−URT−2 turns the UART into the Feetech half−duplex
servo bus and passes the 24 V on to the servos.
UART on module USART2: PA2 TX → pin 4,
pin 3 → PA3 RX (5 V−tolerant). R36 / R37 10k
pull−ups keep RX / TX idle HIGH (TX floats in reset).
Set the FE−URT−2 level switch to 3.3 V (not 5 V):
it only sets the UART levels, the servo 24 V is
unchanged. At 5 V its TX exceeds PA3's limit while
the module is unpowered and back−feeds the 3V3 via R36.
F9 3 A PTC on the servo 24 V. F10 1.1 A PTC on the
FE−URT−2 5 V, fed from the screen converter so a
servo−side short can't pull down the module supply.
24 V only for 24 V servo types.
Blank chip: the ST bootloader probes PA2 / PA3, so flash
over SWD first (the normal flow).

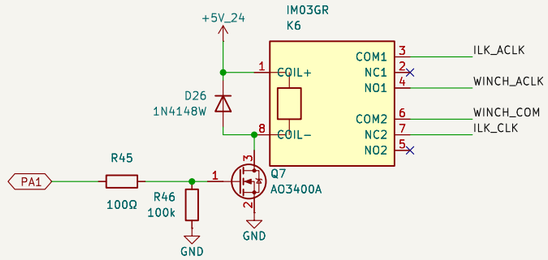
Sheet 9: IMU 9 V supply (as on the MES3.1 carrier) + screen 5 V USB−C power
Aaron Raj Bhatti
Monkey Robotics
Sheet: /AuxPower/
File: AuxPower.kicad_sch
Title: MES4−Carrier−EC

Size: A3	Date: 2026−10−27	**Rev: A**
KiCad E.D.A. 9.0.2		Id: 9/11

Winch direction relays (as U1 / U2 on the MES2_Extension)







Notes

Same relays and terminal as the MES2_Extension "WINCH CONTROL": TE IM03GR, 5 V coil (178 Ω, 28 mA) from +5V_24, contacts 2 A / 220 VDC, 60 W DC. They only switch the 48 V winch controller's direction inputs to J18 pin 2 (COM): pin 1 = CLK, pin 3 = ACLK. They are not rated for the winch motor current.
TE terminal assignment: coil 1 (+) / 8 (−); pole 1: 3 COM, 2 NC, 4 NO; pole 2: 6 COM, 7 NC, 5 NO.

Drivers as on the relay sheets (CIF §7a): AO3400A, 100 Ω gate, 100k pull-down, so both relays are OFF during reset and boot (PA0 / PA1 float then). The Extension used 2SC2884 transistors without a pull-down.

Hardware interlock (new, uses the spare second pole): CLK closes COM → ILK_CLK → K5 NO1 → CLK only while K6 rests (its NC2 feeds ILK_CLK), and the same the other way. If both relays are energised, both outputs stay open, so the winch can never see both directions at once. Firmware still drives only one.

Isolation from the 48 V winch system: J18 and the WINCH_ / ILK_ nets touch only relay contacts. No carrier GND, supply or ESD clamp connects to that side, so the relays' coil-to-contact insulation is the barrier (TE IM series, up to 2500 Vrms). PCB: net class Winch_48V; keep ≥ 2 mm from all other copper and no GND pour under J18 or the contact pins. Never add a TVS / ESD diode to GND on J18: it bridges the isolation.

PA0 / PA1 are relay 1 / 2 coil feedback in the MES4-CIF §2c table. This board drops the coil feedback to free them for the winch (E-Crawler GPIO budget).

Sheet 10: winch direction relays (IM03GR, as on the MES2_Extension) with hardware interlock
Aaron Raj Bhatti
Monkey Robotics
Sheet: /Winch/
File: Winch.kicad_sch
Title: MES4-Carrier-EC
Size: A3 | Date: 2026-10-27 | **Rev: A**
KiCad E.D.A. 9.0.2 | Id: 10/11

J1 FINGER MAP — what every MES4-MCU finger does on this carrier

Finger	Port	Net / signal	Function	Connected to (this carrier)	Carrier parts	MES4-CIF §2c	Notes
SMALL FINGER GROUP 1-11 · CAN + debug							
1	–	CANH	CAN	DB9 pin 3, RJ45 pin 1 (all 4 ports)	J3, J4, ESD D6 / D7	CANH	Transceiver + 120 Ω terminator (JP1) are on the module
3	–	CANL	CAN	DB9 pin 5, RJ45 pin 2	as CANH	CANL	No 24 V on any CAN connector
4	–	+3V3	POWER	M.2 4 → +3V3_MOD: INA236s, ADS1232 DVDD, PC817s, pull-ups	LED2 green, J2 VTref	+3V3 out	≤ 50 mA allowed; this carrier draws ≈ 7 mA worst case
6	PA9	VCP_TX	UART	M.2 6 → ST-Link T_VCP_TX (STDC14 pin 14)	J2, H5VSD3B ESD	VCP_TX	USART1 TX: ST-Link VCP + bootloader UART
7	PA13	SWDIO	DEBUG	M.2 7 → ST-Link SWDIO (STDC14 pin 4)	J2, H5VSD3B ESD	SWDIO	Do not reuse as GPIO
8	PA10	VCP_RX	UART	M.2 8 ← ST-Link T_VCP_RX (STDC14 pin 13)	J2, H5VSD3B ESD	VCP_RX	USART1 RX
9	PA14	SWCLK	DEBUG	M.2 9 → ST-Link SWCLK (STDC14 pin 6)	J2, H5VSD3B ESD	SWCLK	Shared with BOOT0, ignored by the module: no BOOT button
11	PF2	NRST	RESET	ST-Link NRST (STDC14 pin 12) + reset button	J2, SW1	NRST	Pull LOW only; no capacitor or pull-up here (100 nF on the module)
2, 5, 10	–	GND	GND	GND plane	–	GND	
LARGE FINGER GROUP 20-75 · power + GPIO							
20, 22	–	+5V	POWER	Module supply from +5V_24	F2 1.1 A PTC + FB1 ferrite pi filter	+5V in	Module window 4.75-5.5 V
23	PA12	USB_DP	USB	Not connected	–	not used	CIF rule 8: leave open
25	PA11	USB_DN	USB	Not connected	–	not used	CIF rule 8: leave open
29	PC6	PC6	GPIO	Relay 2 drive, LED DRL (out)	Q3 AO3400A, R12 100 Ω, R13 100k	Relay CH2 drive	
31	PA8	PA8	GPIO	Relay 1 drive, safety light (out)	Q2 AO3400A, R9 100 Ω, R10 100k	Relay CH1 drive	May read LOW in reset (UCPD pull-down) until the UCPD1_STROBE write (FW)
35	PA7	PA7	GPIO	Relay 4 drive, spare (out)	Q5 AO3400A, R20 100 Ω, R21 100k	Relay CH4 drive	Bootloader pulls it LOW on a blank chip: relay stays off
37	PA6	PA6	GPIO	ADS1232 PDWN (out, 100k pull-down)	U8 ADS1232	ADS1232 PDWN	ADC held off in reset
39	PA5	PA5	GPIO	ADS1232 SCLK (out, 100k pull-down)	U8 ADS1232	ADS1232 SCLK	TT pin: 3.6 V max
41	PA4	PA4	GPIO	ADS1232 DRDY / DOUT (in, 100 Ω)	U8 ADS1232	ADS1232 DRDY	TT pin: 3.6 V max (DVDD = 3V3). Bit-banged read: DOUT is not on SPI1 MISO
43	PA3	PA3	UART	USART2 RX ← J17 pin 3, door servo (FE-URT-2)	J17, R38 1k, R36 10k pull-up, D23	Relay CH4 coil FB	Differs from CIF. FE-URT-2 must be set to 3.3 V levels
45	PA2	PA2	UART	USART2 TX → J17 pin 4, door servo (FE-URT-2)	J17, R39 100 Ω, R37 10k pull-up, D24	Relay CH3 coil FB	Differs from CIF. Bootloader probes PA2 / PA3 on a blank chip: flash over SWD
47	PA1	PA1	GPIO	Winch ACLK relay drive (out)	K6 IM03GR, Q7 AO3400A, R45 100 Ω, R46 100k	Relay CH2 coil FB	Differs from CIF. Drive only one of PA0 / PA1
49	PA0	PA0	GPIO	Winch CLK relay drive (out)	K5 IM03GR, Q6 AO3400A, R43 100 Ω, R44 100k	Relay CH1 coil FB	Differs from CIF. Both winch relays off in reset (100k pull-downs)
51	PC15	PC15	GPIO	Autonics sensor 2 (in, HIGH = sensor on)	U7 PC817, R27 10k pull-down	Spare input	Weak pin: input only, never source. EXTI15 shared with PA15
53	PB9	I2C1_SDA	I2C	I²C1 SDA: 4 × INA236 (0x40 / 0x41 / 0x42 / 0x43)	R3 2.2k pull-up to +3V3_MOD	I²C1 SDA	400 kHz
55	PB8	I2C1_SCL	I2C	I²C1 SCL: 4 × INA236	R4 2.2k pull-up to +3V3_MOD	I²C1 SCL	
59	PB6	PB6	GPIO	Autonics sensor 1 (in, HIGH = sensor on)	U6 PC817, R25 10k pull-down	Spare input	Bootloader pull-up on a blank chip: reads LOW with the 10k, harmless
61	PB3	PB3	GPIO	Relay 3 drive, battery stopper (out)	Q4 AO3400A, R17 100 Ω, R18 100k	Relay CH3 drive	
63	PA15	PA15	GPIO	INA236 ALERT, all four wire-OR (in)	R5 10k pull-up to +3V3_MOD	INA236 ALERT	EXTI15 shared with PC15: interrupt on PC15, poll ALERT (latched)
others	–	GND	GND	GND plane + socket hold-down pads	–	GND	21, 24, 26-28, 30, 32-34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56-58, 60, 62, 64-75

Totals: all 14 module GPIOs used (10 GPIO + USART2 on PA2 / PA3) + I2C1 + VCP UART + SWD + NRST; CAN transceiver on the module; USB not connected.
Function colours as on the MES4-MCU U1 pin map. Yellow MES4-CIF §2c cells: this board differs from the standard (no relay coil feedback, so PA0 / PA1 = winch, PA2 / PA3 = servo UART).
Checked against the MES4-MCU netlist on 2026-10-27: every finger carries the same signal on both boards, and every GPIO finger goes straight to the STM32G0B1.

Sheet 11: MES4-MCU module pinout and what this carrier connects to each finger
Aaron Raj Bhatti
Monkey Robotics
Sheet: /Pinout/
File: Pinout.kicad_sch
Title: MES4-Carrier-EC
Size: A3 | Date: 2026-10-27 | **Rev: A**
KiCad E.D.A. 9.0.2 | Id: 11/11